Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems

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Abstract

Confluence asks whether reductions from one source rejoin. Many reconstruction problems ask a different question: whether states with the same observable boundary settle to the same public output. This paper gives certificates for that stronger comparison after silent representation changes are quotiented away. Two moduli turn residual and observation errors into an endpoint bound. The bound survives arbitrary refinement depth, and compatible finite levels define a solver-independent projective limit. The paper also proves repair and symmetry obstructions, gives a checkable conditional-expectation resampling projector, and classifies two succinct decision problems as coNP- and Π_2^P-complete. For a declared finite scheduler, four nested protected-behavior tests separate empty fibers, failed positive reach, non-almost-sure reach, and ambiguous first-hit endpoint classes, and an exact finite-state morphism preserves the resulting profile. On the canonical OPH adaptive repair branch, every attempt stream is eventually constant; pathwise weak fairness makes the stable state normal, while work conservation gives a normal horizon bounded by the initial mismatch count. Under the separately stated completeness and confluence hypotheses, weak-fair runs reach one canonical raw and public endpoint. Counting every scheduler invocation, including stutter, gives a genuinely restrictive cumulative capacity: no initial-mismatch-only finite attempt horizon exists for arbitrary normalizing schedulers, whereas bounded waste gives a quantitative upper horizon, with an exact sharp threshold in the work-conserving case, and a capacity-bounded public endpoint. On the separate input-independent compiler branch, one emission-order round-robin cycle reaches the exact output in n attempts, every accepted repair path is quadratically bounded and a same-metric family makes the worst case order-sharp $\Theta(n^2)$, and weak fairness alone has no uniform finite attempt horizon. The Markov-chain and fair-termination ingredients are standard; the contribution is their source-bound assembly around protected observations, canonical repair, and the existing behavior-cut interface. These results supply the observable-normal-form machinery used by the Observer-Patch Holography consensus and Yang–Mills papers.

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1 Introduction

A confluent rewrite system has a familiar geometry: paths may split, but paths from the same source eventually rejoin. In many applications the source itself is hidden. What is fixed is an interface, input, conserved sector, sensor record, or boundary condition. The relevant question is then not

$$x \to^* y, \quad x \to^* z \quad \Longrightarrow \quad y \downarrow z,$$

but rather

$$B(x) = B(x') \quad \Longrightarrow \quad \text{the settled observable outputs of } x \text{ and } x' \text{ agree.}$$

The two statements have different proof obligations. Confluence is a property of paths. Observable reconstruction is a property of the consistent set and of the observation map on that set.

We use *observation-determined normal form* for a settled state determined through an observation map on the consistent set. This is distinct from the control-theory use of "observable normal form" for a state-space form obtained by coordinate transformation, including in Boolean control networks [13].

Let Q be a state space after quotienting representation changes that are regarded as silent, let $C \subseteq Q$ be the consistent states, and let

$$B : Q \longrightarrow \mathcal{B}$$

be protected observable data. For $b \in \mathcal{B}$, set

$$C_b := C \cap B^{-1}(b).$$

The three possibilities

$$|C_b| = 0, \qquad |C_b| = 1, \qquad |C_b| > 1$$

mean, respectively, that the observation is unrealizable, reconstructing, or ambiguous. Termination and confluence do not alter this trichotomy. They control how a computation reaches C; they do not determine how many points of C share one observation.

The distinction becomes more consequential under perturbation. Exact injectivity of $B|_C$ says nothing about conditioning. A sequence of finite systems may have injective observation maps at every stage and nevertheless become arbitrarily unstable under refinement. To capture the missing information, we use two moduli:

(i) an error-bound modulus carrying a constraint residual to distance from C;

(ii) an inverse-observation modulus carrying observation discrepancy within C to state discrepancy.

The first is an error-bound or metric-subregularity object in the sense of variational analysis [15, 19, 27, 36]. The second is the modulus of continuity of the inverse of $B|_C$. Both notions are standard. Their use as an *observation-determined normal-form certificate* for constraint and rewrite systems yields the uniform and multi-stage refinement calculus developed here.

1.1 Main contributions

The manuscript has six technical contributions.

C1. Correct exact interface. We state a canonical partial normalizer and a boundary-indexed unique-normal-form theorem with the empty-fiber case separated from weak normalization. An audited-terminal theorem further separates computational quiescence from semantic consistency. Identification modulo a silent equivalence is characterized exactly by cross-source endpoint agreement. For a separately declared finite scheduler, we also distinguish the static fiber question from positive first hit, almost-sure first hit, and endpoint-class selection. The exact confluence implications are explicitly treated as classical.

C2. Sharp observational stability. For a residual Φ and an observation map B, we define intrinsic moduli η_Φ and ω_B. For arbitrary residual levels δ_x, δ_y,

$$\begin{aligned} d_Q(x,y) \le{}& \eta_\Phi(\delta_x) + \eta_\Phi(\delta_y) \\ &+ \omega_B(d_{\mathcal{B}}(Bx,By) + L_B[\eta_\Phi(\delta_x) + \eta_\Phi(\delta_y)])\,. \end{aligned} \tag{1.1}$$

At zero residual the bound is exact by definition. Across all finite metric systems, the coefficient 2 in the equal-residual specialization cannot be lowered.

C3. Uniform refinement towers. One-step approximate naturality follows from consistency and observation defects. We distinguish same-level receipt-conditioned agreement from genuinely cross-level comparison. An anchor-optimized theorem compares independently computed outputs at arbitrary refinement levels after restriction to a common level. Compatible nested inputs leave only the intervening defect segment, and summable pathwise defects yield an implementation-independent projective limit. A counterexample shows that stagewise injectivity is not enough without a uniform inverse modulus.

C4. Repair and selection obstructions. A total exact local repair preserving an exterior collar exists if and only if the local consistency relation projects surjectively onto that collar. For a finite scheduler, four nested protected-behavior sets and their complementary cuts isolate fiber, positive-reach, almost-sure, and endpoint-selection obstructions. On the canonical OPH adaptive branch, every attempt stream stabilizes after finitely many genuine changes, pathwise weak fairness is sufficient for normality, and work conservation gives an initial-mismatch bound on the normal horizon. Counting stutter as positive attempt cost separates that bound from cumulative capacity: no single mismatch-only attempt horizon uniformly bounds all normalizing schedulers, bounded waste gives a quantitative upper horizon, and two nonisomorphic canonical finite sources make the work-conserving threshold sharp. Completeness and confluence remain explicit requirements for a consistent canonical endpoint. The underlying finite Markov, weighted-rewriting, and fair-termination facts are standard; the contribution is the source-bound protected-observation certificate, the adaptive attempt/cost classification, and their exact connection to the existing behavior-cut and public-endpoint interfaces. On a finite full-support probability space, observation-fiber resampling is the weighted orthogonal conditional-expectation projector and has a finite matrix recognition receipt. A stabilizer obstruction prevents an equivariant selector on ambiguous fibers.

C5. Constructive and linear classes. Ranked functional-audit systems have singleton-or-empty fibers, terminate under every effective asynchronous schedule with an explicit update

bound, and settle synchronously in dependency depth; acyclic Boolean circuits compile directly into this class. Linear quotient systems admit rank tests and singular-value bounds that instantiate (1.1); finite linear dynamics yield the usual observability matrix as a special case.

C6. Receipts and complexity. A standard finite Markov drift lemma turns a row-wise kernel inequality into endpoint, stationary, and expected-occupation receipts, without asserting infinite-horizon confinement under persistent noise. Machine-checked first-hit laws, exact finite-state transport, and native TwoBit, fixed-relation, and proper-target counterexamples provide separate receipts for the protected-behavior profile. For succinct Boolean systems, we record tight coNP- and Π_2^P-completeness classifications in this formulation; these supporting results are not part of the principal novelty claim.

1.2 Relation to existing theory

The exact theory overlaps several established areas. Abstract rewriting supplies Newman's lemma, Church–Rosser properties, unique normal forms, and normalization theory [3, 8, 28, 43, 49]. Rewriting modulo equivalence and confluence under invariants address quotient and relevant-state restrictions [12, 14, 24, 38]. Quantitative rewriting enriches rewrite relations themselves and proves metric analogues of classical confluence theorems [25]. Error bounds and inverse stability are mature subjects in variational analysis [15, 36]. Functoriality of rewriting is also established in categorical graph transformation [21].

Cumulative and nonuniform rewrite costs, derivation-height bounds, failed-attempt accounting, and self-stabilizing enabled-move convergence are established theory [7, 18, 34]. The contribution below is not a new weighted-rewriting formalism. It locates the quantitative boundary on the declared chosen-site attempt semantics, where a scheduler may spend capacity on a quiescent site, and then proves the exact additional condition and finite OPH witnesses that recover a sharp bound.

The triangle inequality, error bounds, inverse continuity, section criteria, boundary-indexed unique normal forms, and the Lipschitz telescope are standard. The integrated certificate architecture combines them with an explicit anchor-optimized cross-level receipt comparison and its cofinal-limit consequence. Approximate inverse-system theory supplies the neighboring framework, while the result here is an organizing synthesis for automated reasoning, rewriting, and semantics.

2 Relation to existing theories

2.1 Rewriting, invariants, and quotients

The classical properties of an abstract reduction system are standard [8, 28, 49]. Confluence restricted to an invariant or taken modulo an equivalence has been developed in Constraint Handling Rules and graph transformation [12, 14, 24]. Those frameworks are close to the first arrow in

$$\text{presentations} \longrightarrow Q \longrightarrow \mathcal{B} :$$

they identify states that should count as the same or restrict attention to relevant states. Our observation map is the second arrow. It may be strictly coarser than equality in Q, and it is required to be injective only on the consistent subset when exact reconstruction is desired.

Unique-normal-form properties weaker than confluence are well known and have been automated for term rewriting [3]. The fiber trichotomy and partial-normalizer universal property here are ele-

mentary set theory; only the reachable-normal-form and fiberwise-confluence corollaries specialize classical rewriting theory.

2.2 Quantitative rewriting versus output stability

Gavazzo and Di Florio enrich rewriting relations over quantales and derive quantitative Newman, Church–Rosser, and critical-pair theorems [25]. Here the rewrite relation remains ordinary. Distances are attached to two semantic interfaces: residual-to-solution error and inversion of observable data on the solution set. Thus the theory does not estimate a quantitative rewrite path; it estimates disagreement between settled outputs of possibly unrelated schedules or initial interiors.

2.3 Error bounds and inverse mappings

The inequality

$$\operatorname{dist}(x, C) \leq \eta(\Phi(x))$$

is an error-bound property. In variational analysis, local linear versions are equivalent to metric subregularity or calmness of suitable set-valued mappings [15, 19, 36]. The inverse-observation modulus is a global finite or compact analogue of inverse Lipschitz/calmness information. Our use of these notions is intentionally transparent: the novelty claim concerns their operational consequences for schedules and refinement towers, not their analytic origin.

2.4 Approximate inverse systems

Approximate inverse systems and almost-commuting diagrams have a substantial topological literature, including stability and resolution theorems for compacta [39–41]. Those works perturb or relax compatibility of bonding data in topological or cover terms. Our setting keeps the ambient inverse system exact and instead gives pointwise metric bounds for normalizer defects, solver receipts, and protected-observation mismatch. The connection limits the novelty claim: the contribution is an explicit certificate adapted to observation-determined normalizers, not the invention of approximate inverse systems.

2.5 Static reconstruction and functional observability

Control-theoretic observability reconstructs a state, or a prescribed function of state, from an output history. Linear observability and functional observability supply rank and conditioning tests; nonlinear work characterizes when a nonlinear functional can be reconstructed from measurements and studies the quality of the inverse map [31, 35, 42]. This is the closest neighboring language for the map $B|_C$. Our setting is nevertheless static at the theorem interface: C is the settled consistent set, B may be any protected quotient observation, and the questions are how residual error and inverse conditioning control schedule disagreement and how these certificates propagate through refinement towers.

2.6 Constraints, databases, and global sections

Acyclic database schemes and positive-semiring relations provide local-to-global consistency results [6, 10]. Constraint satisfaction studies local consistency as an algorithmic reduction [16]. Database repair semantics distinguishes the existence of many consistent repairs from the answers invariant across them [5]. Sheaf-theoretic contextuality similarly turns failure of a global section into an

obstruction [2]. Our empty-fiber certificate belongs to this broad lineage. The additional layer is the interaction with normal-form stability and refinement.

2.7 Fair scheduling and asynchronous Boolean-network fixing

Fairness is a qualitative liveness condition rather than a latency bound; Lamport formulates the general closure issue explicitly [37], while the self-stabilization daemon taxonomy separates fairness, boundedness, and enabledness as independent scheduling axes [20]. Asynchronous Boolean-network work studies fixing words and their minimal lengths, including monotone, acyclic, tree, and conjunctive families [4, 23]. We assign no novelty credit to weak fairness, round robin, fixing words, or quadratic fixing phenomena themselves. The source-specific contribution below is the joint machine-checked classification for the actual nonmonotone `go` compiler: its first emission-order cycle, taken in the order in which the compiler emits nodes, is the existing sweep and therefore has a linear exact-output horizon, its adversarial genuine accepted-step paths are order-sharp quadratic in the same emitted-node metric, and its Tower endpoint distinguishes both from arbitrary weak-fair attempts and from a separately supplied bounded-waste contract. General well-formed federations remain sharply exponential, in line with the order of the acyclic fixing-length results [23]; the source-specific delta is that the compiler's single-consumer shape is exactly what removes the exponent.

2.8 Asynchronous functional networks

Kahn process networks provide timing-independent determinacy for networks of functional components, while generalized Kahn principles treat broader abstract asynchronous, including nondeterministic, models under their stated hypotheses [1, 30]. Ranked functional systems below are a finite terminating special case with residual audits. Their purpose is to give a class in which the exact boundary hypotheses, update bound, and obstruction output are explicit. They do not reprove those principles.

2.9 Finite Markov reachability and lumping

First-hitting probabilities, the decomposition into transient and closed classes in a finite chain, least nonnegative solutions of reachability Bellman equations, and strong lumpability are standard [9, 32, 44]. Markov functions and intertwining provide neighboring language for law pushforward [46]. We use these facts without a novelty claim. Their role here is architectural: declared active sources and a protected observation index four separate failure modes; those modes recover the pre-existing OPH behavior cuts through explicit supported traces; and exact transport retains the active-source, target, observation, and silent-quotient data in addition to the kernel law.

3 Exact observable fibers

Definition 3.1 (Observable quotient system)**.** An *observable quotient system* is a tuple

$$\mathfrak{S} = (Q, C, \mathcal{B}, B),$$

where Q is a set, $C \subseteq Q$ is the consistent subset, $\mathcal{B}$ is an observation set, and $B : Q \to \mathcal{B}$ is the protected observation map. In applications, Q may itself be a quotient Σ/Γ of presentations by gauge, implementation, or representation changes.

Definition 3.2 (Fiber trichotomy)**.** For $b \in \mathcal{B}$, define $C_b = C \cap B^{-1}(b)$. The observation b is *unrealizable*, *reconstructing*, or *ambiguous* according as $|C_b|$ is 0, 1, or greater than 1. The system is *observable-determined* if $B|_C$ is injective.

Definition 3.3 (Canonical partial normalizer)**.** If $B|_C$ is injective, define

$$\mathsf{N}_B : Q \longrightarrow C \sqcup \{\bot\}$$

by returning the unique member of $C_{B(x)}$ when the fiber is nonempty and $\bot$ otherwise. The symbol $\bot$ denotes a mathematically proved no-extension outcome, not a failure of search.

Remark 3.4 (Proof-carrying interpretation)*.* For finite data with decidable membership and equality, exhaustive search returns the dependent alternative

$$\left(\sum_{c \in C} [B(c) = B(x)]\right) \;\sqcup\; [C_{B(x)} = \varnothing].$$

Thus "obstructed" carries a proof of empty fiber. In the ordinary classical setting the same alternative is defined by excluded middle and choice, but injectivity alone is not an algorithm for deciding which side holds. Even for finite succinct inputs, such a proof need not be computationally cheap; Section 9 shows that realizability is NP-complete.

Theorem 3.5 (Universal property)**.** *The following are equivalent.*

(a) $B|_C$ *is injective.*

(b) There is a unique map $N : Q \to C \sqcup \{\bot\}$ *such that:*

(i) $N(x) = c \in C$ *implies* $B(c) = B(x)$*;*
(ii) $N(x) = \bot$ *if and only if* $C_{B(x)} = \varnothing$*;*
(iii) $N(c) = c$ *for all* $c \in C$*;*
(iv) $B(x) = B(y)$ *implies* $N(x) = N(y)$*.*

(c) $B|_C : C \to B(C)$ *is a bijection.*

If $B(Q) \subseteq B(C)$*, then* N *is a total idempotent retraction* $Q \to C$ *preserving* B*.*

Proof. If $B|_C$ is injective, every nonempty fiber contains exactly one point, so N_B has all listed properties and any competing map is forced to agree with it fiber by fiber. Conversely, if $c, c' \in C$ have $B(c) = B(c')$, then boundary extensionality and the fixed-point property give $c = N(c) = N(c') = c'$. The equivalence with (c) is immediate. □

Let $\to$ be a relation on Q, write $\to^*$ for its reflexive transitive closure, and let $\mathrm{NF}(\to)$ be the normal forms.

Definition 3.6. The relation $\to$ is *observation preserving* if $x \to y$ implies $B(x) = B(y)$, and *complete for* C if $\mathrm{NF}(\to) = C$.

Proposition 3.7 (Reachable normal forms live in the observable fiber)**.** *If* $\to$ *preserves* B *and is complete for* C*, then every normal form reachable from* x *belongs to* $C_{B(x)}$*.*

Proof. Completeness puts the reachable normal form in C, and observation preservation puts it in the fiber of $B(x)$. □

Theorem 3.8 (Cross-source endpoint identification modulo equivalence)**.** *Assume that $\to$ preserves B and that $\mathrm{NF}(\to) = C$. Let $\approx$ be an equivalence relation on C. The following are equivalent:*

(i) for all $c, d \in C$, $B(c) = B(d)$ implies $c \approx d$;

(ii) whenever $B(x) = B(y)$, $x \to^* n_x$, $y \to^* n_y$, and $n_x, n_y \in \mathrm{NF}(\to)$, one has $n_x \approx n_y$.

Thus observation identifiability modulo $\approx$ is exactly agreement modulo $\approx$ of reachable endpoints from possibly different *sources with the same observation.*

Proof. Under (i), Proposition 3.7 puts n_x, n_y in C and observation preservation gives $B(n_x) = B(x) = B(y) = B(n_y)$, hence $n_x \approx n_y$. Conversely, take $c, d \in C$ with $B(c) = B(d)$. Since $C = \mathrm{NF}(\to)$ and $\to^*$ is reflexive, apply (ii) to $c \to^* c$ and $d \to^* d$. □

Corollary 3.9 (Silent-quotient criterion)**.** *In addition to the hypotheses of Theorem 3.8, suppose $c \approx d$ implies $B(c) = B(d)$. Then $B|_C$ induces a surjection*

$$\overline{B} : C/{\approx} \longrightarrow B(C), \qquad [c] \longmapsto B(c),$$

and $\overline{B}$ is injective, hence bijective, if and only if either condition of Theorem 3.8 holds. This includes quotients by silent representation or gauge changes; equality recovers ordinary endpoint uniqueness.

Proof. Silence makes $\overline{B}$ well defined and it is surjective by its codomain. Its injectivity says exactly that $B(c) = B(d)$ implies $[c] = [d]$, equivalently $c \approx d$. □

Remark 3.10 (Three separate hypotheses). Same-source confluence compares arms of a peak from one state; Theorem 3.8 compares endpoints from all sources in one observation fiber. Weak normalization is a separate existence hypothesis ensuring that every source has at least one endpoint. Strong normalization, fairness, or another liveness argument is needed to conclude that every allowed schedule settles. None of these three hypotheses may be substituted for either of the others.

Remark 3.11 (Static fibers versus a stochastic scheduler). The fiber C_b is determined by (Q, C, B) alone. Statements that a source reaches C_b with positive probability, hits it almost surely, or selects one endpoint class require additional data: a transition kernel, a nonempty set of declared active sources, and an equivalence on first-hit endpoints. None of those dynamic conclusions follows from singleton fibers or confluence alone; conversely, a stochastic reachability statement does not construct a local repair with prescribed write support.

Corollary 3.12 (Correct empty/singleton alternative)**.** *Under the hypotheses of Proposition 3.7:*

(i) if $C_b = \varnothing$, no state of $B^{-1}(b)$ reaches a normal form;

(ii) if $C_b = \{c_b\}$, every reachable normal form from $B^{-1}(b)$ equals c_b;

(iii) if, in addition, every state of $B^{-1}(b)$ is weakly normalizing, then every such state reduces to c_b, and $\to$ is confluent on that fiber.

Proof. The first two claims follow directly from Proposition 3.7. For (iii), normalize both arms of any peak. Their normal forms are both c_b, providing a join. □

The completeness hypothesis $\mathrm{NF}(\to) = C$ is sometimes too strong: a computation may halt uniquely and then fail a separate semantic audit. The next theorem separates those two layers.

Theorem 3.13 (Audited terminal alternative)**.** *Let $T = \mathrm{NF}(\to)$ be the computational terminal set and let $C \subseteq T$ be the semantically consistent terminals. Assume that $\to$ preserves B, every state is weakly normalizing, and $B|_T$ is injective. Then every $x \in Q$ reaches a unique terminal $t_x \in T$, and*

$$B(t_x) = B(x), \qquad t_x \in C \iff C_{B(x)} \neq \varnothing.$$

Thus a realizable observation has a unique reachable terminal at its consistent extension, whereas an unrealizable observation can have a unique reachable terminal only outside C. The theorem does not assert strong normalization of every execution.

Proof. Weak normalization supplies a reachable $t_x \in T$, and observation preservation gives $B(t_x) = B(x)$. Any other reachable terminal has the same observation, so injectivity of $B|_T$ makes it equal to t_x. If $t_x \in C$, it witnesses a nonempty consistent fiber. Conversely, if $c \in C_{B(x)}$, then $c \in T$ and $B(c) = B(t_x)$; injectivity on T gives $c = t_x$, hence $t_x \in C$. □

When $T = C$, Theorem 3.13 reduces to the realizable part of Corollary 3.12. When $T \supsetneq C$, it prevents an audit failure from being misreported as nontermination or as a consistent normal form.

Remark 3.14 (Classical scope). Part (iii) is the invariant-fiber form of the classical implication from weak normalization and uniqueness of normal forms with respect to reduction to confluence. It identifies the exact hypotheses needed for observable reconstruction.

4 Observational stability moduli

Let (Q, d_Q) be a finite metric space, let $(\mathcal{B}, d_{\mathcal{B}})$ be any metric space, let $\varnothing \neq C \subseteq Q$, and let $\Phi : Q \to \mathbb{R}_{\geq 0}$ satisfy

$$C = \Phi^{-1}(0).$$

All maxima below therefore exist. The same formulas extend to compact spaces by replacing maxima with suprema and imposing the continuity hypotheses stated in Proposition 4.13; the displayed Lipschitz estimates continue to assume a finite Lipschitz constant for B.

Definition 4.1 (Residual error-bound modulus)**.** Define

$$\eta_\Phi(t) := \max\{\mathrm{dist}_Q(x, C) : \Phi(x) \leq t\}.$$

An *error-bound modulus* is any nondecreasing η satisfying

$$\mathrm{dist}_Q(x, C) \leq \eta(\Phi(x)).$$

Definition 4.2 (Inverse-observation modulus)**.** Define

$$\omega_B(r) := \max\{d_Q(c, c') : c, c' \in C,\ d_{\mathcal{B}}(Bc, Bc') \leq r\}.$$

The system is *stably observable-determined* if $\omega_B(r) \to 0$ as $r \downarrow 0$.

Proposition 4.3 (Minimality of the intrinsic moduli)**.** *The functions η_Φ and ω_B are nondecreasing. The following also hold:*

(i) $\eta_\Phi(0) = 0$*;*

(ii) η_Φ *is the least nondecreasing error-bound modulus on the attained residual values;*

(iii) $\omega_B(0) = 0$ *if and only if* $B|_C$ *is injective;*

(iv) ω_B *is the least nondecreasing modulus satisfying*

$$d_Q(c, c') \le \omega(d_{\mathcal{B}}(Bc, Bc')) \qquad (c, c' \in C).$$

For finite C*, injectivity of* $B|_C$ *implies that there is an* $r_0 > 0$ *such that* $\omega_B(r) = 0$ *for* $0 \le r < r_0$*. When* $|C| > 1$*, one may take* r_0 *to be the minimum distance between distinct points of* $B(C)$*; when* $|C| = 1$*, any positive* r_0 *works.*

Proof. Monotonicity follows from inclusion of the defining sets. Since $C = \Phi^{-1}(0)$, residual zero has distance zero from C. If η is any nondecreasing error-bound modulus and $\Phi(x) \le t$, then $\operatorname{dist}(x, C) \le \eta(\Phi(x)) \le \eta(t)$; taking the maximum gives $\eta_\Phi(t) \le \eta(t)$. The proof for ω_B is identical. The statement at zero is exactly injectivity, and finiteness gives a positive minimum separation among distinct boundary images. □

Let L_B be any Lipschitz constant for B on Q.

Theorem 4.4 (Heterogeneous two-output estimate)**.** *If* $x, y \in Q$ *satisfy*

$$\Phi(x) \le \delta_x, \qquad \Phi(y) \le \delta_y, \qquad d_{\mathcal{B}}(Bx, By) \le \varepsilon,$$

then

$$\begin{aligned} d_Q(x, y) \le{}& \eta_\Phi(\delta_x) + \eta_\Phi(\delta_y) \\ &+ \omega_B(\varepsilon + L_B[\eta_\Phi(\delta_x) + \eta_\Phi(\delta_y)])\,. \end{aligned} \tag{4.1}$$

Proof. Choose nearest points $c_x, c_y \in C$. Then

$$d_Q(x, c_x) \le \eta_\Phi(\delta_x), \qquad d_Q(y, c_y) \le \eta_\Phi(\delta_y).$$

Lipschitz continuity and the triangle inequality give

$$\begin{aligned} d_{\mathcal{B}}(Bc_x, Bc_y) &\le d_{\mathcal{B}}(Bc_x, Bx) + d_{\mathcal{B}}(Bx, By) + d_{\mathcal{B}}(By, Bc_y) \\ &\le \varepsilon + L_B[\eta_\Phi(\delta_x) + \eta_\Phi(\delta_y)]. \end{aligned}$$

The definition of ω_B bounds $d_Q(c_x, c_y)$. A final triangle inequality yields (4.1). □

The symmetric form used most often is

$$d_Q(x, y) \le 2\eta_\Phi(\delta) + \omega_B(\varepsilon + 2L_B\eta_\Phi(\delta))\,. \tag{4.2}$$

Corollary 4.5 (Rate transfer)**.** *Suppose, on the residual and observation ranges under consideration,*

$$\eta_\Phi(t) \le at^p, \qquad \omega_B(r) \le br^q$$

for constants $a, b \ge 0$ *and exponents* $p, q > 0$*. Then*

$$\begin{aligned} d_Q(x, y) \le{}& a(\delta_x^p + \delta_y^p) \\ &+ b\Big(\varepsilon + L_B a(\delta_x^p + \delta_y^p)\Big)^q \end{aligned}$$

under the hypotheses of Theorem 4.4. In particular, linear error bounds and a Lipschitz inverse give a linear output certificate.

Proof. Substitute the two rate bounds into (4.1). □

4.1 Optimality and the exact observational modulus

Definition 4.6 (Settled-output modulus)**.** For $\delta, r \geq 0$, define

$$\Theta_{\Phi,B}(\delta, r) := \max\{d_Q(x, y) : \Phi(x), \Phi(y) \leq \delta,\ d_{\mathcal{B}}(Bx, By) \leq r\}.$$

Theorem 4.7 (Sharpness)**.** *The following statements hold.*

(i) At zero residual,

$$\Theta_{\Phi,B}(0, r) = \omega_B(r).$$

Thus the inverse-observation term in (4.2) is exact.

(ii) No universal estimate of the form

$$\Theta_{\Phi,B}(\delta, 0) \leq a\,\eta_\Phi(\delta) + \omega_B(0)$$

can hold for all finite metric systems with a constant $a < 2$.

Proof. When $\delta = 0$, the admissible points are precisely C, so the two defining maxima coincide.

For (ii), fix $s > 0$ and let $Q = \{c, x, y\}$ with distances

$$d(c, x) = d(c, y) = s, \qquad d(x, y) = 2s.$$

Let $C = \{c\}$, let B be constant, and set $\Phi(c) = 0$, $\Phi(x) = \Phi(y) = 1$. Then $\eta_\Phi(1) = s$, $\omega_B(0) = 0$, and $\Theta_{\Phi,B}(1, 0) = 2s$. □

Remark 4.8. The theorem does not assert that every term in (4.1) can be attained simultaneously in every structured class. It states the correct universal boundary: the observation term is forced on C, and the coefficient 2 on two independent residual errors is globally sharp.

4.2 Approximate schedule independence

A finite execution is a sequence $x = x_0 \to x_1 \to \cdots \to x_n$. It is (δ, β)-*settled* if its endpoint z satisfies

$$\Phi(z) \leq \delta, \qquad d_{\mathcal{B}}(Bz, Bx) \leq \beta.$$

No exact observation-preservation assumption is made.

Corollary 4.9 (Approximate schedule independence)**.** *Suppose two executions start at the same state x and end at z_1, z_2, where the ith execution is (δ_i, β_i)-settled. Then*

$$\begin{aligned} d_Q(z_1, z_2) \leq\ & \eta_\Phi(\delta_1) + \eta_\Phi(\delta_2) \\ & + \omega_B(\beta_1 + \beta_2 + L_B[\eta_\Phi(\delta_1) + \eta_\Phi(\delta_2)]). \end{aligned} \tag{4.3}$$

The same bound holds for executions starting from two different states with equal initial observations.

Proof. The endpoint observation discrepancy is at most $\beta_1 + \beta_2$. Apply Theorem 4.4. □

Corollary 4.10 (High-probability schedules)**.** *Let Z_1, Z_2 be random endpoints of two schedules from a common initial state in finite Q. Suppose schedule i is (δ_i, β_i)-settled with probability at least $1 - p_i$. Let D denote the right-hand side of (4.3). Then*

$$\mathbb{P}[d_Q(Z_1, Z_2) \leq D] \geq 1 - p_1 - p_2$$

and

$$\mathbb{E}[d_Q(Z_1, Z_2)] \leq D + (p_1 + p_2)\,\mathrm{diam}(Q).$$

Proof. Use the union bound for the two settling events. On their intersection apply Corollary 4.9; on its complement use the diameter bound. □

The preceding result consumes a probabilistic settling receipt. The following standard drift calculation is one finite mechanism for producing such a receipt.

Proposition 4.11 (Finite Markov drift receipt)**.** *Let K be a Markov kernel on finite Q, write*

$$(Kf)(x) = \sum_{y \in Q} K(x,y) f(y),$$

and let $V : Q \to \mathbb{R}_{\geq 0}$. Suppose $0 \leq \kappa < 1$, $\xi \geq 0$, and

$$(KV)(x) \leq \kappa V(x) + \xi \qquad (x \in Q).$$

For the chain (X_n) started at x,

$$\mathbb{E}_x[V(X_n)] \leq \kappa^n V(x) + \xi \frac{1-\kappa^n}{1-\kappa}.$$

Consequently, for every $\delta > 0$,

$$\mathbb{P}_x[V(X_n) > \delta] \leq \frac{\kappa^n V(x) + \xi(1-\kappa^n)/(1-\kappa)}{\delta}.$$

If $V = \Phi$ and every transition with positive probability preserves B, then the complementary event supplies a $(\delta, 0)$-settling receipt for Corollary 4.10. When the kernel is meant to sample rewrite schedules, its support must additionally consist of permitted rewrite steps or stuttering moves. This is a one-time endpoint bound; pathwise confinement over a time interval requires an additional excursion or concentration hypothesis.

Proof. The tower property and the drift inequality give $a_{n+1} \leq \kappa a_n + \xi$ for $a_n = \mathbb{E}_x[V(X_n)]$. Induction solves this scalar recurrence and gives the displayed geometric sum. The probability bound is Markov's inequality. Positive-probability transitions preserve the initial observation along every sampled path. □

Corollary 4.12 (Stationary and occupation receipts)**.** *Under the hypotheses of Proposition 4.11, every stationary distribution π satisfies, for $\delta > 0$,*

$$\mathbb{E}_\pi V \leq \frac{\xi}{1-\kappa}, \qquad \pi\{V > \delta\} \leq \frac{\xi}{(1-\kappa)\delta}.$$

For a chain started at x and $N \geq 1$,

$$\mathbb{E}_x\left[\frac{1}{N}\sum_{n=0}^{N-1} \mathbf{1}_{\{V(X_n)>\delta\}}\right] \leq \frac{V(x)}{N\delta(1-\kappa)} + \frac{\xi}{\delta(1-\kappa)}.$$

The same right-hand side bounds the probability of being outside the δ-tube at an independent uniformly sampled time in $\{0, \ldots, N-1\}$. When $\xi > 0$, these expectation and occupation bounds do not imply infinite-horizon pathwise confinement; persistent noise can permit repeated excursions.

Proof. Integrate the drift inequality against π and use stationarity. For the occupation bound, use $\mathbf{1}_{\{V>\delta\}} \leq V/\delta$, sum the estimate in Proposition 4.11, and bound $\sum_{n=0}^{N-1} \kappa^n \leq (1-\kappa)^{-1}$. □

4.3 Compact and uniform families

Proposition 4.13 (Compact-space extension)**.** *Let Q be compact metric, let $C \subseteq Q$ be nonempty and closed, let $\Phi : Q \to \mathbb{R}_{\geq 0}$ be continuous with $C = \Phi^{-1}(0)$, and let $B : Q \to \mathcal{B}$ be continuous. Then $\eta_\Phi(t) \to 0$ as $t \downarrow 0$. If $B|_C$ is injective, then its inverse on $B(C)$ is uniformly continuous and $\omega_B(r) \to 0$ as $r \downarrow 0$.*

Proof. If $\eta_\Phi(t)$ did not tend to zero, there would be $t_n \downarrow 0$ and x_n satisfying $\Phi(x_n) \leq t_n$ and $\operatorname{dist}(x_n, C) \geq a > 0$. A convergent subsequence has limit x with $\Phi(x) = 0$, hence $x \in C$, contradicting the distance bound. The second statement follows because a continuous bijection from compact C to the Hausdorff space $B(C)$ is a homeomorphism, whose inverse is uniformly continuous. □

Let $\{\mathfrak{S}_i\}_{i \in I}$ be a nonempty family of compact or finite metric systems with a common Lipschitz bound L for the maps B_i. Define

$$H(t) := \sup_i \eta_i(t), \qquad \Omega(r) := \sup_i \omega_i(r).$$

We assume these suprema are finite on the ranges under consideration; this uniform boundedness is not automatic when the diameters of the systems are unbounded.

A *uniform settled-output modulus* is a function $\Psi(t, r)$ such that $\Theta_{\Phi_i, B_i}(t, r) \leq \Psi(t, r)$ for every $i \in I$.

Theorem 4.14 (Uniform observational stability)**.** *If $H(t) \to 0$ and $\Omega(r) \to 0$ at zero, then every member of the family satisfies the uniform estimate*

$$d_i(x, y) \leq 2H(\delta) + \Omega(r + 2LH(\delta))$$

whenever $\Phi_i(x), \Phi_i(y) \leq \delta$ and $d_{\mathcal{B}_i}(B_i x, B_i y) \leq r$. Conversely, any uniform settled-output modulus restricts at zero residual to a modulus dominating Ω.

Proof. Apply Theorem 4.4 stagewise and take suprema. At zero residual, Theorem 4.7(i) gives the converse statement. □

Example 4.15 (Stagewise uniqueness without uniform stability)**.** For $n \geq 1$, let $C_n = Q_n = \{0, 1\}$ with discrete state distance, and let $B_n(0) = 0$, $B_n(1) = 1/n$ in $\mathbb{R}$. Every B_n is injective, yet for every $r > 0$ one has $\sup_n \omega_{B_n}(r) = 1$. Thus exact uniqueness at every finite stage does not supply a stable refinement limit.

Example 4.16 (Stagewise error bounds without a uniform error bound)**.** For $n \geq 1$, let $Q_n = \{0, 1\}$ with the discrete metric, let $C_n = \{0\}$, let B_n be constant, and put

$$\Phi_n(0) = 0, \qquad \Phi_n(1) = 1/n.$$

For every fixed n, $\eta_n(t) \to 0$ as $t \downarrow 0$, and the unique consistent fiber is perfectly conditioned. Nevertheless, for every $t > 0$ one has $\sup_n \eta_n(t) = 1$. Thus uniform residual control and uniform inverse-observation control are independent requirements.

Executable uniformity controls. The accompanying finite proof-obligation certificate stores descriptions of both infinite counterfamilies and an executable witness constructor: for every positive integer m, choose stage $n = m$ and radius or residual threshold $1/m$. The exact witness has observation or residual gap $1/m$ and state error one. Since $1/m \to 0$, the quantified examples above, rather than any fixed finite prefix, prove that the family moduli do not vanish. The receipt regression-checks exact instances and records the cross-multiplied null-sequence proof schema.

Further controls describe the infinite unit-defect tower $p_j = j, a_j = 1$, whose tail from n to $n+M$ is M, and a restriction map that amplifies fine distance $1/8$ to coarse distance 1 when its Lipschitz receipt is removed. A three-point parametric family independently shows why the observation maps need one common Lipschitz bound: their stagewise constants grow like n, while residuals and observation gaps vanish and the state gap tends to one. The cofinal-limit corollary has separate controls for incomplete coordinate spaces, incompatible inputs, and nonvanishing solver errors. The arbitrary-depth positive telescope is the mathematical theorem above and its Lean counterpart; the family-uniform modulus and cofinal-limit quantifiers are not claimed as Lean formalizations.

4.4 Parallel composition and observation design

Theorem 4.17 (Exact product calculus)**.** *Let I be nonempty and finite. For each $i \in I$, let*

$$\mathfrak{S}_i = (Q_i, C_i, \mathcal{B}_i, B_i, \Phi_i)$$

be a finite metric system with $C_i \neq \varnothing$. Form the parallel product

$$Q = \prod_{i\in I} Q_i, \quad C = \prod_{i\in I} C_i, \quad B = (B_i)_{i\in I}, \quad \Phi(x) = \max_i \Phi_i(x_i),$$

with maximum metrics on Q and $\prod_i \mathcal{B}_i$. Then

$$\eta_\Phi(t) = \max_i \eta_{\Phi_i}(t), \qquad \omega_B(r) = \max_i \omega_{B_i}(r).$$

Consequently, parallel composition preserves a uniform stability certificate without a dimension-dependent loss in the maximum metric.

Proof. For the maximum product metric,

$$\operatorname{dist}_Q(x, C) = \max_i \operatorname{dist}_{Q_i}(x_i, C_i).$$

The constraint $\Phi(x) \le t$ is equivalent to $\Phi_i(x_i) \le t$ for every i. Taking maxima gives the upper bound for η_Φ; equality is attained by choosing a maximizing component and arbitrary consistent points in all other components. The proof for ω_B is identical: the product observation discrepancy is at most r exactly when every component discrepancy is at most r. □

Corollary 4.18 (Adding observations cannot worsen conditioning)**.** *Let $D : C \to \mathcal{D}$ be an additional observation and equip $\mathcal{B} \times \mathcal{D}$ with the maximum metric. For the enriched map $B^+ = (B, D)$,*

$$\omega_{B^+}(r) \le \omega_B(r).$$

Thus extra quotient-visible sensors can only improve the intrinsic inverse-observation certificate; they may, of course, carry their own implementation error outside C.

Proof. The pairs of consistent states satisfying $d_{\mathcal{B}\times\mathcal{D}}(B^+c, B^+c') \le r$ form a subset of those satisfying $d_{\mathcal{B}}(Bc, Bc') \le r$. □

5 Naturality and refinement towers

For an observable system define its reconstructing, realizable domain by

$$Q^{\mathrm{adm}} := \{q \in Q : \text{there exists a unique } c \in C \text{ with } B(c) = B(q)\}.$$

Every normalizer in this section is understood to act on a chosen metric subspace of Q^{adm}, with the consistent set restricted to the corresponding normalizer image; every coarse map is assumed to preserve the chosen admissible subspaces. To avoid carrying restriction notation through every formula, we relabel those subspaces and restricted consistent sets as Q_s, Q_r, Q_n and C_s, C_r, C_n. Thus $B_n|_{C_n}$ is injective on the declared domain. Obstructed and ambiguous fibers are governed by Section 3; they are not silently assigned a finite normalizer value in the metric estimates below.

Let

$$\mathfrak{S}_s = (Q_s, C_s, \mathcal{B}_s, B_s, \Phi_s), \qquad \mathfrak{S}_r = (Q_r, C_r, \mathcal{B}_r, B_r, \Phi_r)$$

be finite or compact observable systems on the admissible domains declared above. Their normalizers N_s and N_r are therefore total on these domains. Let $f : Q_s \to Q_r$ be a coarse state map and $h : \mathcal{B}_s \to \mathcal{B}_r$ a coarse observation map.

Definition 5.1 (Morphism defects)**.** The pair (f, h) has observation defect ε_B and consistency defect δ_C if

$$d_{\mathcal{B}_r}(B_r(fq), h(B_s q)) \le \varepsilon_B \qquad (q \in Q_s), \tag{5.1}$$

$$\Phi_r(fc) \le \delta_C \qquad (c \in C_s). \tag{5.2}$$

Theorem 5.2 (One-step approximate naturality)**.** *Assume B_r is L_r-Lipschitz. Under (5.1)–(5.2),*

$$d_{Q_r}(fN_s q, N_r fq) \le \eta_r(\delta_C) + \omega_r(2\varepsilon_B + L_r\eta_r(\delta_C)) \tag{5.3}$$

for every relevant $q \in Q_s$.

Proof. Put $y = fN_s q$ and $z = N_r fq$. By (5.2), $\Phi_r(y) \le \delta_C$. Choose $c \in C_r$ with $d_r(y, c) \le \eta_r(\delta_C)$. Since both normalizers preserve observations,

$$\begin{aligned} d_{\mathcal{B}_r}(B_r c, B_r z) &\le d_{\mathcal{B}_r}(B_r c, B_r y) + d_{\mathcal{B}_r}(B_r fN_s q, hB_s N_s q) \\ &\quad + d_{\mathcal{B}_r}(hB_s q, B_r fq) \\ &\le L_r\eta_r(\delta_C) + 2\varepsilon_B. \end{aligned}$$

Both c and z lie in C_r, so the inverse-observation modulus bounds $d_r(c, z)$. Add $d_r(y, c)$. □

Corollary 5.3 (Exact naturality from uniqueness)**.** *If $f(C_s) \subseteq C_r$ and $B_r \circ f = h \circ B_s$, then*

$$f \circ N_s = N_r \circ f.$$

Proof. Both sides are consistent coarse states with the same coarse observation. Reconstructing fibers make them equal. Equivalently, set both defects to zero in Theorem 5.2. □

Corollary 5.4 (Stability under perturbation of the consistency model)**.** *Let (Q, d_Q) carry two nonempty compact consistent sets C and $\widetilde{C}$, two observation maps $B, \widetilde{B} : Q \to \mathcal{B}$, and total maps $N, \widetilde{N} : Q \to Q$ satisfying*

$$N(Q) \subseteq C, \qquad \widetilde{N}(Q) \subseteq \widetilde{C}, \qquad B \circ N = B, \qquad \widetilde{B} \circ \widetilde{N} = \widetilde{B}.$$

Suppose

$$d_H(C, \widetilde{C}) \le \alpha, \qquad \sup_{q \in Q} d_{\mathcal{B}}(Bq, \widetilde{B}q) \le \beta,$$

where d_H is Hausdorff distance. If B is L-Lipschitz and $B|_C$ has inverse modulus ω_B, then

$$d_Q(Nx, \widetilde{N}x) \le \alpha + \omega_B(2\beta + L\alpha) \qquad (x \in Q).$$

Proof. Choose $c \in C$ with $d_Q(c, \widetilde{N}x) \le \alpha$. Since N and $\widetilde{N}$ preserve their respective observations,

$$\begin{aligned} d_{\mathcal{B}}(Bc, BNx) &\le d_{\mathcal{B}}(Bc, B\widetilde{N}x) + d_{\mathcal{B}}(B\widetilde{N}x, \widetilde{B}\widetilde{N}x) \\ &\quad + d_{\mathcal{B}}(\widetilde{B}x, Bx) \\ &\le L\alpha + 2\beta. \end{aligned}$$

Both c and Nx lie in C, so $d_Q(c, Nx) \le \omega_B(L\alpha + 2\beta)$. Add the Hausdorff error. □

Remark 5.5. Categorical functoriality of rewrite steps is an established subject [21]. Corollary 5.3 is a different and simpler statement: it concerns the terminal normalizer, and uniqueness allows it to be proved without simulating each fine rewrite. The nontrivial content for applications is the defect estimate, its accumulation across a tower, and the perturbation certificate of Corollary 5.4.

5.1 Accumulation through arbitrary depth

Let

$$Q_0 \xleftarrow{\rho_{1,0}} Q_1 \xleftarrow{\rho_{2,1}} Q_2 \leftarrow \cdots$$

be an inverse sequence of nonempty metric spaces with Lipschitz restriction maps, with $\rho_{m,n} = \rho_{n+1,n} \circ \cdots \circ \rho_{m,m-1}$. Let $N_n : Q_n \to Q_n$ be normalizers. This subsection deliberately restarts with general metric spaces; the finite-or-compact ambient assumption used for the preceding uniform-family examples is not retained. Completeness is imposed only where a Cauchy estimate is converted into a coordinate limit. Assume the following pathwise and uniform one-step normalization defects are finite:

$$\alpha_n(q) := d_n(\rho_{n+1,n} N_{n+1} q, N_n \rho_{n+1,n} q), \qquad a_n := \sup_{q \in Q_{n+1}} \alpha_n(q).$$

For $j \ge n$, write

$$K_{j,n} := \mathrm{Lip}(\rho_{j,n}), \qquad K_{n,n} = 1.$$

Theorem 5.6 (Telescoping refinement bound)**.** *For every $m > n$ and every $q \in Q_m$,*

$$d_n(\rho_{m,n} N_m q, N_n \rho_{m,n} q) \le \sum_{j=n}^{m-1} K_{j,n} a_j. \tag{5.4}$$

Proof. For $j = n, \dots, m$, define

$$p_j := \rho_{j,n} N_j \rho_{m,j} q.$$

Then $p_m = \rho_{m,n} N_m q$ and $p_n = N_n \rho_{m,n} q$. For $j < m$, put $r = \rho_{m,j+1} q$. The difference between consecutive terms is

$$\begin{aligned} d_n(p_{j+1}, p_j) &= d_n(\rho_{j,n} \rho_{j+1,j} N_{j+1} r, \rho_{j,n} N_j \rho_{j+1,j} r) \\ &\le K_{j,n} a_j. \end{aligned}$$

Summing the telescoping chain proves (5.4). □

Corollary 5.7 (Summable-defect asymptotic naturality)**.** *If all restriction maps are nonexpansive, then*

$$d_n(\rho_{m,n}N_m q, N_n\rho_{m,n}q) \le \sum_{j=n}^{m-1} a_j.$$

If $\sum_j a_j < \infty$, the defect is uniformly small when the comparison level tends outward along the refinement index. More generally, the sufficient condition is

$$\sup_{m>n} \sum_{j=n}^{m-1} K_{j,n}a_j \longrightarrow 0 \quad (n \to \infty).$$

Combining Theorem 5.2 with Theorem 5.6 gives an explicit tower certificate in terms of residual, observation, and Lipschitz data at each stage.

Corollary 5.8 (Modulus-generated tower certificate)**.** *Suppose the nth restriction map has consistency defect δ_n and observation defect ε_n, and let*

$$a_n^* := \eta_n(\delta_n) + \omega_n(2\varepsilon_n + L_n\eta_n(\delta_n))\,.$$

Then (5.4) holds with a_j replaced by a_j^*.

Definition 5.9 (Run-specific approximate solver receipt)**.** At level m, an output $\widehat{x}_m \in Q_m$ produced from $q_m \in Q_m$ has receipt (δ_m, β_m) when

$$\Phi_m(\widehat{x}_m) \le \delta_m, \qquad d_{\mathcal{B}_m}(B_m\widehat{x}_m, B_m q_m) \le \beta_m.$$

Set

$$e_m(\delta, \beta) := \eta_m(\delta) + \omega_m(\beta + L_m\eta_m(\delta))\,.$$

Uniform receipts for a solver map mean that the two inequalities hold for every admissible input, but the comparison results below require only the displayed run-specific evidence.

Lemma 5.10 (Receipt-to-exact comparison)**.** *If $\widehat{x}_m$ has receipt (δ_m, β_m) relative to q_m, then*

$$d_m(\widehat{x}_m, N_m q_m) \le e_m(\delta_m, \beta_m). \tag{5.5}$$

Proof. The exact normal form has residual zero and preserves the observation of q_m. Apply the heterogeneous two-output estimate to $\widehat{x}_m$ and $N_m q_m$, using $\eta_m(0) = 0$. □

Proposition 5.11 (Same-level implementation agreement)**.** *Let $n \le m$. If $\widehat{x}_m$ and $\widehat{y}_m$ are outputs on the same input q_m with receipts (δ_m, β_m) and (δ'_m, β'_m), then*

$$d_n(\rho_{m,n}\widehat{x}_m, \rho_{m,n}\widehat{y}_m) \le K_{m,n}(e_m(\delta_m, \beta_m) + e_m(\delta'_m, \beta'_m)).$$

No refinement-defect term occurs in this same-level comparison.

Proof. Compare both outputs with the same exact state $N_m q_m$ using Lemma 5.10, then apply the $K_{m,n}$-Lipschitz restriction map. □

For $n \le h \le r$ and $q_r \in Q_r$, define the amplified pathwise defect segment

$$\mathsf{A}_{r,h}^{(n)}(q_r) := \sum_{j=h}^{r-1} K_{j,n}\,\alpha_j(\rho_{r,j+1}q_r)\,,$$

with an empty sum equal to zero. By definition of a_j,

$$\mathsf{A}_{r,h}^{(n)}(q_r) \le \sum_{j=h}^{r-1} K_{j,n}a_j.$$

Lemma 5.12 (Pathwise telescoping to an anchor)**.** *For $n \le h \le r$ and $q_r \in Q_r$,*

$$d_n(\rho_{r,n}N_r q_r, \rho_{h,n}N_h\rho_{r,h}q_r) \le \mathsf{A}^{(n)}_{r,h}(q_r).$$

Proof. For $j = h, \dots, r$, put $p_j = \rho_{j,n}N_j\rho_{r,j}q_r$. Consecutive terms satisfy

$$d_n(p_{j+1}, p_j) \le K_{j,n}\alpha_j(\rho_{r,j+1}q_r).$$

Summing these inequalities proves the claim. □

Corollary 5.13 (Fine-to-coarse solver certificate)**.** *For $m \ge n$ and an output $\widehat{x}_m$ with receipt (δ_m, β_m) relative to q_m,*

$$d_n(\rho_{m,n}\widehat{x}_m, N_n\rho_{m,n}q_m) \le K_{m,n}e_m(\delta_m, \beta_m) + \mathsf{A}^{(n)}_{m,n}(q_m).$$

Proof. Use Lemma 5.10, apply $\rho_{m,n}$, and then use Lemma 5.12 with $h = n$. □

Theorem 5.14 (Anchored cross-level solver comparison)**.** *Let $n \le h \le \min\{m, \ell\}$. Let $q_m \in Q_m$ and $q_\ell \in Q_\ell$, and suppose $\widehat{x}_m$ and $\widehat{y}_\ell$ have receipts (δ_m, β_m) and $(\delta'_\ell, \beta'_\ell)$, respectively. Define the protected observation mismatch at the anchor by*

$$\gamma_h := d_{\mathcal{B}_h}(B_h\rho_{m,h}q_m, B_h\rho_{\ell,h}q_\ell)\,.$$

Then

$$\begin{aligned} &d_n(\rho_{m,n}\widehat{x}_m, \rho_{\ell,n}\widehat{y}_\ell) \\ &\quad\le K_{m,n}e_m(\delta_m, \beta_m) + \mathsf{A}^{(n)}_{m,h}(q_m) + K_{h,n}\omega_h(\gamma_h) + \mathsf{A}^{(n)}_{\ell,h}(q_\ell) + K_{\ell,n}e_\ell(\delta'_\ell, \beta'_\ell). \end{aligned} \tag{5.6}$$

The two pathwise segments may be replaced by their uniform a_j bounds. Because the estimate holds for every admissible h, the right-hand side may be minimized over the possible anchors.

Proof. Compare each approximate output with its exact normal form using Lemma 5.10, and apply the corresponding restriction to level n. Use Lemma 5.12 to move both exact branches to level h. The two anchor normal forms are consistent, preserve the observations of their anchor inputs, and hence differ in Q_h by at most $\omega_h(\gamma_h)$. Applying $\rho_{h,n}$ contributes $K_{h,n}$. The triangle inequality gives (5.6). □

Corollary 5.15 (Nested compatible solver levels)**.** *Under the output and receipt hypotheses of Theorem 5.14, let $n \le m \le \ell$ and suppose*

$$B_m q_m = B_m\rho_{\ell,m}q_\ell.$$

Then

$$\begin{aligned} d_n(\rho_{m,n}\widehat{x}_m, \rho_{\ell,n}\widehat{y}_\ell) \le{}& K_{m,n}e_m(\delta_m, \beta_m) + \mathsf{A}^{(n)}_{\ell,m}(q_\ell) \\ &+ K_{\ell,n}e_\ell(\delta'_\ell, \beta'_\ell). \end{aligned}$$

In particular, the middle term is at most $\sum_{j=m}^{\ell-1} K_{j,n}a_j$. The compatibility hypothesis holds when $q_m = \rho_{\ell,m}q_\ell$.

Proof. Take $h = m$ in Theorem 5.14. The first pathwise segment is empty and $\omega_m(0) = 0$ on the observation-determined consistent domain. □

Corollary 5.16 (Cofinal projective implementation independence)**.** *Assume every Q_n is complete and every restriction map is nonexpansive. Let $q = (q_n) \in \varprojlim Q_n$ be a compatible admissible input and suppose its pathwise exact-normalizer defects are summable:*

$$\sum_{j=0}^{\infty} \alpha_j(q_{j+1}) < \infty.$$

For $i \in \{1, 2\}$, let $\widehat{x}_m^{(i)}$ be outputs at q_m whose receipt errors

$$e_m^{(i)} := e_m(\delta_m^{(i)}, \beta_m^{(i)}) \longrightarrow 0.$$

Then, for every fixed n, the limits

$$x_n = \lim_{m \to \infty} \rho_{m,n} \widehat{x}_m^{(i)}$$

exist, are independent of i, and are unchanged by passage to a cofinal subsequence. The family $x = (x_n)$ belongs to $\varprojlim Q_n$, and

$$d_n(x_n, N_n q_n) \le \sum_{j=n}^{\infty} \alpha_j(q_{j+1}).$$

If all pathwise normalizer defects vanish, then $x_n = N_n q_n$ for every n.

Proof. For $s \ge r \ge n$, compatibility gives $q_r = \rho_{s,r} q_s$. Corollary 5.15 and nonexpansiveness give, for either pair of implementations,

$$d_n\Big(\rho_{r,n} \widehat{x}_r^{(i)}, \rho_{s,n} \widehat{x}_s^{(i')}\Big) \le e_r^{(i)} + e_s^{(i')} + \sum_{j=r}^{s-1} \alpha_j(q_{j+1}).$$

The right-hand side tends to zero as $r, s \to \infty$. Completeness gives the coordinate limits, and the same estimate makes them independent of the implementation and of a cofinal subsequence. Continuity of the restriction maps passes compatibility to the limits. Finally, Corollary 5.13 gives

$$d_n\Big(\rho_{m,n} \widehat{x}_m^{(i)}, N_n q_n\Big) \le e_m^{(i)} + \sum_{j=n}^{m-1} \alpha_j(q_{j+1});$$

let $m \to \infty$. □

The uniform hypotheses used earlier imply $e_m^{(i)} \to 0$ whenever $\delta_m^{(i)}, \beta_m^{(i)} \to 0$, the observation maps have a common Lipschitz bound, and $H(t), \Omega(t) \to 0$ at zero. Uniform summability of $\sum_j a_j$ implies the pathwise summability condition. Thus raw receipt parameters alone are not enough: uniform conditioning and the amplified restriction factors are load-bearing.

5.2 Exact inverse limits

Assume that all one-step defects vanish and that $\rho_{n+1,n}(C_{n+1}) \subseteq C_n$. Let

$$Q_\infty := \varprojlim Q_n = \{(q_n) : \rho_{n+1,n} q_{n+1} = q_n\},$$

and similarly let $C_\infty = \varprojlim C_n$. Here Q_∞ is the inverse limit of the admissible domains fixed at the start of the section. It is the full ambient inverse limit only when every observation value is reconstructing and realizable.

Theorem 5.17 (Inverse-limit normalizer)**.** *For each n, let $N_n : Q_n \to C_n \subseteq Q_n$ be an idempotent observation-preserving normalizer that fixes every point of C_n. Assume the restriction maps form an inverse system, preserve the consistent subsets, and satisfy*

$$\rho_{n+1,n} N_{n+1} = N_n \rho_{n+1,n} \qquad (n \geq 0).$$

Assume also that the observation spaces carry compatible restriction maps $\beta_{n+1,n} : \mathcal{B}_{n+1} \to \mathcal{B}_n$ with

$$B_n \rho_{n+1,n} = \beta_{n+1,n} B_{n+1}.$$

Then

$$N_\infty((q_n)) = (N_n q_n)$$

defines an idempotent map $Q_\infty \to C_\infty$ preserving the induced inverse-limit observation. If, for every n and every $b_n \in B_n(C_n)$, the fiber $C_n \cap B_n^{-1}(b_n)$ is a singleton, then every realizable inverse-limit consistent observation fiber is a singleton.

Proof. Naturality makes $(N_n q_n)$ a compatible family. Because each coordinate lies in C_n and each N_n is idempotent, the resulting map lands in C_∞ and is idempotent. Compatibility of the B_n makes observation preservation coordinatewise. If two inverse-limit consistent families have the same inverse-limit observation, their nth coordinates lie in the same finite consistent observation fiber for every n, hence agree at every coordinate. □

Remark 5.18 (What the theorem does not prove). The theorem supplies a normalizer on a declared inverse system. It does not prove compactness, nonemptiness, a continuum dynamics, or adequacy of the chosen refinement maps. Those are separate mathematical or model-specific premises.

6 Repairability and selection obstructions

The existence of a unique consistent state with a given observation does not imply that a prescribed local write support can construct it. This section separates extension, uniqueness, and repair support.

Let X_W be an interior write space, let X_D be a protected collar space, and let

$$R \subseteq X_W \times X_D$$

be the locally consistent relation.

Definition 6.1 (Strong local repair)**.** A *strong local repair* is a map

$$\mathsf{R} : X_W \times X_D \longrightarrow R$$

such that

(i) $\pi_D \mathsf{R}(x, d) = d$ for every (x, d);

(ii) $\mathsf{R}(r) = r$ for every $r \in R$.

Thus every local state is repaired, the collar is preserved, and consistent states are fixed.

Theorem 6.2 (Collar-section criterion)**.** *For nonempty finite X_W and finite X_D, a strong local repair exists if and only if the collar projection*

$$\pi_D : R \longrightarrow X_D$$

is surjective.

Proof. A repair sends every (x, d) to a point of R with collar d, so surjectivity is necessary. Conversely, choose a section $s : X_D \to R$ of π_D and define

$$\mathsf{R}(z) = \begin{cases} z, & z \in R, \\ s(\pi_D z), & z \notin R. \end{cases}$$

This is a strong repair. □

If $R = \varnothing$ while $X_W \times X_D \neq \varnothing$, nonexistence of a repair is immediate; when the domain is empty, the unique empty map is a vacuous repair. The next proposition assumes $R \neq \varnothing$ because ordinary real-valued distance to the empty set is convention-dependent; an extended-distance formulation may instead assign that case margin $+\infty$.

Corollary 6.3 (Finite no-repair certificate)**.** *For nonempty X_W, a single collar value $d \notin \pi_D(R)$ certifies that no total exact repair preserving that collar exists.*

Proposition 6.4 (Stable no-repair margin)**.** *Suppose X_W and X_D are metric spaces, $\varnothing \neq R \subseteq X_W \times X_D$, and the product metric dominates collar distance:*

$$d_{X_W \times X_D}((w, d), (w', d')) \geq d_{X_D}(d, d').$$

For $d \in X_D$, set

$$\mu_R(d) := \operatorname{dist}_{X_D}(d, \pi_D(R)).$$

Then for every $w \in X_W$,

$$\operatorname{dist}_{X_W \times X_D}((w, d), R) \geq \mu_R(d).$$

If R is compact and $d \notin \pi_D(R)$, then $\mu_R(d) > 0$. Hence no procedure whose output preserves the collar value d exactly can approximate consistency to error below $\mu_R(d)$.

Proof. For every $(w', d') \in R$, the product-metric assumption gives

$$d((w, d), (w', d')) \geq d_{X_D}(d, d') \geq \mu_R(d).$$

Take the infimum. Compactness makes $\pi_D(R)$ compact and therefore closed, so a point outside it has positive distance. □

Definition 6.5 (Repair width)**.** Let $\mathscr{W}$ be a finite family of admissible write supports with a cost function $w : \mathscr{W} \to \mathbb{R}_{\geq 0}$. For each $W \in \mathscr{W}$, let $R_W \subseteq X_W \times X_{D(W)}$ be the induced local consistency relation and let $D_{\mathrm{adm}}(W) \subseteq X_{D(W)}$ be the declared admissible collar domain. The *repair width* is

$$\min\{w(W) : D_{\mathrm{adm}}(W) \subseteq \pi_{D(W)}(R_W)\},$$

with value $+\infty$ when no admissible support has the required projection property.

The definition prevents a one-site no-go from being misreported as a global inconsistency: increasing the transaction support may restore surjectivity.

6.1 A protected-behavior first-hit profile

This section adds stochastic data without identifying it with the static repair question. Let Q be finite and let $K : Q \times Q \to [0,1]$ be a row-stochastic kernel. Fix a protected carrier $P \subseteq \mathcal{B}$. For every $b \in P$, let $I_b \subseteq Q$ be a nonempty set of declared active sources and put $T_b = C_b$. Let $\approx_b$ be an equivalence relation on T_b. The first entrance time

$$\tau_b := \inf\{n \geq 0 : X_n \in T_b\}$$

includes time zero. For $q \in I_b$ and $c \in T_b$, define the endpoint mass

$$h_b(q,c) := \Pr_q[X_{\tau_b} = c,\ \tau_b < \infty]$$

and its total $h_b(q) = \sum_{c \in T_b} h_b(q,c)$. These are determined by the increasing finite-horizon sums of first-hit path weights; no arbitrary initial probability law is part of the qualitative definition.

Definition 6.6 (Protected-behavior layers and cuts)**.** Define nested subsets of P by

$$\begin{aligned}
L_0 &= \{b \in P : T_b \neq \varnothing\},\\
L_1 &= \{b \in L_0 : \exists q \in I_b,\ h_b(q) > 0\},\\
L_2 &= \{b \in L_1 : \forall q \in I_b,\ h_b(q) = 1\},\\
L_3 &= \left\{ b \in L_2 : \begin{array}{l} h_b(q,c) > 0,\ h_b(q',c') > 0 \\ \Longrightarrow c \approx_b c' \quad (q, q' \in I_b) \end{array} \right\}.
\end{aligned}$$

The cumulative cuts are $K_j = P \setminus L_j$. The disjoint obstruction strata are

$$\Delta_0 = K_0, \qquad \Delta_j = K_j \setminus K_{j-1} \quad (1 \leq j \leq 3).$$

Thus Δ_0 is an empty-fiber obstruction, Δ_1 a failure of any declared source to supply positive reach, Δ_2 a failure of all-source almost-sure reach, and Δ_3 a failure of endpoint-class selection.

Theorem 6.7 (Profile partition and structural characterizations)**.** *The layers satisfy*

$$L_3 \subseteq L_2 \subseteq L_1 \subseteq L_0 \subseteq P,$$

and

$$P = \Delta_0 \sqcup \Delta_1 \sqcup \Delta_2 \sqcup \Delta_3 \sqcup L_3.$$

Moreover:

(i) $b \in L_1$ *exactly when a positive-support finite path from some* $q \in I_b$ *first enters* T_b *at its endpoint;*

(ii) $b \in L_2$ *exactly when* $b \in L_1$ *and no active source can reach, before hitting* T_b*, a nonempty closed positive-support set disjoint from* T_b*;*

(iii) $b \in L_3$ *exactly when* $b \in L_2$ *and the union of positive first-hit endpoint supports over* I_b *is one* $\approx_b$*-class.*

Proof. Nesting is immediate from the definitions, after which successive set differences give the displayed disjoint partition. For (i), expand the finite-horizon first-hit mass as the sum of products of transition weights; nonnegativity makes the sum positive exactly when one summand is positive. For (ii), the states reachable before the first hit form a finite directed support graph. A positive probability of avoiding T_b forever reaches a bottom strongly connected component disjoint from T_b, while any reachable closed such component retains positive mass away from the target. This is the standard finite-chain closed-class argument. Part (iii) simply rewrites the pairwise definition in terms of the union of positive endpoint supports. □

The all-source quantifier here means all members of the declared I_b, not all states of Q. Positive reach, almost-sure hit, and endpoint selection are different predicates. In particular, $h_b(q) = 1$ supplies neither a rate nor an expected hitting-time bound, and singleton support modulo $\approx_b$ need not mean literal endpoint equality.

Proposition 6.8 (Exact behavior-cut recovery)**.** *Construct traces from positive-support paths, observe each trace by its protected behavior, and define four admissibility levels respectively by target existence, positive supported reach, absence of a reachable closed trap for every active source, and one endpoint class over all active sources. If support-to-rewrite soundness, rewrite-to-support completeness, and target normality hold, then the four realized-behavior sets are exactly L_0, L_1, L_2, L_3 and the corresponding behavior cuts are exactly K_0, K_1, K_2, K_3.*

Proof. At level zero the trace witness reduces to nonempty target fiber. Level one uses part (i) of Theorem 6.7; level two uses part (ii); and level three uses part (iii). The two scheduler laws identify supported paths with observation-preserving rewrite paths in both directions. Taking complements inside P gives the cut equalities. The structural certificates are explicit parts of this adapter; the proposition does not claim that an arbitrary admissibility profile has these four meanings. □

Consequently, variant comparison is coordinatewise: $(K_0, K_1, K_2, K_3) \leq (K'_0, K'_1, K'_2, K'_3)$ when every coordinate is included. This is a product preorder, not a scalar score or total order.

Theorem 6.9 (Exact finite-state profile transport)**.** *Consider two finite protected profiles and maps $f : Q \to Q'$ and $g : P \to P'$. Suppose protected membership is reflected and covered; active sources are reflected and covered along f; target membership is reflected and every target endpoint is covered; observations commute with f and g; the target equivalences satisfy*

$$c \approx_b d \quad \Longleftrightarrow \quad f(c) \approx'_{g(b)} f(d);$$

and the kernels obey exact lumping

$$K'(f(q), y) = \sum_{x : f(x) = y} K(q, x).$$

Then first-hit endpoint mass pushes forward along f, total hit probability is preserved, and for every $b \in P$ and $0 \leq j \leq 3$,

$$b \in L_j \Longleftrightarrow g(b) \in L'_j, \qquad b \in K_j \Longleftrightarrow g(b) \in K'_j.$$

Proof. Induction on the first-hit time gives the exact endpoint-mass pushforward: the base case uses target reflection and coverage, and the step uses kernel lumping. Monotone limits preserve the equality for total and endpoint hit probabilities. Active-source coverage and target coverage reflect positive mass and almost-sure mass in both directions, while quotient exactness reflects the singleton endpoint-class condition. The four layer equivalences and then their complementary cut equivalences follow. Observation commutation records that the transported target is the same protected behavior; it is not a model-to-code refinement claim. □

6.2 Finite observation-conditioned resampling

There is nevertheless a canonical *averaging* operation once a reference measure is supplied; unlike a section, it does not select one state from an ambiguous fiber.

Theorem 6.10 (Fiber averaging is conditional expectation)**.** *Let Q be finite, let $\mu(x) > 0$ with $\sum_x \mu(x) = 1$, and let $B : Q \to \mathcal{B}$. For $b \in B(Q)$ put $F_b = B^{-1}(b)$ and $\mu(F_b) = \sum_{y \in F_b} \mu(y)$. Define*

$$(P_B f)(x) = \frac{1}{\mu(F_{B(x)})} \sum_{y \in F_{B(x)}} \mu(y) f(y).$$

Then $P_B = \mathbb{E}_\mu[\,\cdot \mid \sigma(B)]$. In the weighted inner product $\langle f, g\rangle_\mu = \sum_x \mu(x) f(x) g(x)$ it satisfies

$$P_B^2 = P_B, \qquad P_B^* = P_B, \qquad \|P_B f\|_{2,\mu} \le \|f\|_{2,\mu},$$

and its fixed space is exactly

$$\mathcal{M}_B = \{f \in \mathbb{R}^Q : B(x) = B(y) \Rightarrow f(x) = f(y)\},$$

the observation-measurable functions.

A Markov matrix K on Q equals the matrix of P_B if and only if the following finite receipt holds:

$$K(x, y) = 0 \qquad \text{when } B(x) \neq B(y), \tag{R1}$$

$$K(x, \cdot) = K(x', \cdot) \qquad \text{when } B(x) = B(x'), \tag{R2}$$

$$\mu(x) K(x, y) = \mu(y) K(y, x) \qquad \text{for all } x, y. \tag{R3}$$

Equivalently, its entries are

$$K(x, y) = \mathbf{1}_{\{B(x) = B(y)\}} \frac{\mu(y)}{\mu(F_{B(x)})}.$$

Proof. The displayed average is constant on each fiber and fixes every fiber-constant function, which proves the conditional-expectation and fixed space statements. Grouping both variables by fibers gives

$$\langle P_B f, g\rangle_\mu = \sum_{b \in B(Q)} \frac{(\sum_{x \in F_b} \mu(x) f(x))(\sum_{y \in F_b} \mu(y) g(y))}{\mu(F_b)},$$

which is symmetric in f, g. Hence P_B is an orthogonal projection and is L^2-contractive. Its matrix plainly satisfies (R1)–(R3). Conversely, (R1)–(R2) give one probability row k_b on each fiber. By (R3), $\mu(x) k_b(y) = \mu(y) k_b(x)$ there, so normalization forces $k_b(y) = \mu(y)/\mu(F_b)$. □

Remark 6.11. Theorem 6.10 recognizes the exact finite resampling projector from checkable matrix identities. It neither chooses a representative of a fiber nor supplies a spectral gap or a convergence rate for a different repair dynamics; those require additional structure. A noncircular receipt enumerates the states and fibers, derives K independently from the declared transition table or transition counts, and then verifies (R1)–(R3); constructing K from the target entry formula is not a verification.

6.3 Symmetry and hidden tie-breaking

Let a group G act on C and $\mathcal{B}$, and assume $B|_C$ is equivariant.

Theorem 6.12 (Equivariant-section criterion)**.** *Let a finite group G act on C and $B(C)$, and let $B|_C : C \to B(C)$ be an equivariant surjection. A G-equivariant section $s : B(C) \to C$ exists if and only if, for one (equivalently every) representative b of each G-orbit in $B(C)$, the fiber C_b contains a point fixed by the stabilizer*

$$G_b = \{g \in G : gb = b\}.$$

Proof. Necessity follows from $gs(b) = s(gb) = s(b)$ for $g \in G_b$. Conversely, choose an orbit representative b and a G_b-fixed point $c_b \in C_b$. Define $s(gb) = gc_b$. If $gb = g'b$, then $g^{-1}g' \in G_b$, so $gc_b = g'c_b$; hence the definition is well posed. It is visibly a section and equivariant. Repeat independently on each orbit. □

Corollary 6.13 (Stabilizer obstruction)**.** *If some fiber C_b has no G_b-fixed point, there is no equivariant boundary-only selector on that orbit. In particular, a transitive ambiguous fiber with no fixed point cannot be canonically collapsed without adding symmetry-breaking data.*

Remark 6.14 (No selection by normalization)*.* Lexicographic order, an energy, a prior, maximum entropy, or an external sector label can select an ambiguous fiber. None is entailed by consistency and the observation alone. The theorem is elementary, but it is a useful firewall: a normal-form map cannot silently become a physical or probabilistic selector without additional structure.

7 Ranked functional-audit systems

We give a broad constructive class in which the partial normalizer, asynchronous dynamics, and obstruction output are all explicit.

Definition 7.1 (Ranked functional-audit system)**.** Let

$$V = V_0 \sqcup V_1 \sqcup \cdots \sqcup V_m$$

be a finite ranked site set with $V_0 \neq \varnothing$. Every site $v \in V$ has a nonempty finite state set X_v, and sites in V_0 are boundary sites. Every generated site $v \in V_k$, $k > 0$, has a parent set

$$P(v) \subseteq V_0 \cup \cdots \cup V_{k-1},$$

and a function

$$F_v : \prod_{u \in P(v)} X_u \longrightarrow X_v.$$

A state satisfies the generating constraints when

$$x_v = F_v(x|_{P(v)}) \qquad (v \notin V_0).$$

A finite family of audits $A_j(x) \in \{0, 1\}$ imposes residual global constraints. A state is globally consistent when it satisfies all generating constraints and every audit returns 1.

Write

$$Q = \prod_{v \in V} X_v, \qquad B : Q \longrightarrow \prod_{v \in V_0} X_v$$

for the full state space and the boundary-coordinate projection. For a boundary value $b \in \prod_{v \in V_0} X_v$, define $E(b)$ recursively by rank.

Theorem 7.2 (Singleton-or-empty fiber)**.** *For every boundary b, exactly one of the following holds:*

(i) every audit passes on $E(b)$ and the consistent fiber is $\{E(b)\}$;

(ii) some audit fails on $E(b)$ and the consistent fiber is empty.

Proof. Rank recursion constructs one state satisfying all generating constraints. Any other such state with boundary b agrees at rank 0 and therefore agrees inductively at every rank. The audits either accept this sole candidate or eliminate it. □

Definition 7.3 (Asynchronous functional update)**.** For $v \notin V_0$, let T_v replace x_v by $F_v(x|_{P(v)})$ and leave every other coordinate unchanged. An update is *effective* when it changes x_v.

Define recursively

$$M(v) = 0 \quad (v \in V_0), \qquad M(v) = 1 + \sum_{u \in P(v)} M(u) \quad (v \notin V_0).$$

Theorem 7.4 (Finite asynchronous normalization with an update bound)**.** *In every sequence of effective asynchronous updates, site v is updated at most $M(v)$ times. Consequently every such sequence has length at most*

$$\sum_{v \in V} M(v),$$

and every maximal sequence terminates at $E(Bx)$ independently of update order.

Proof. Proceed by induction on rank. Boundary sites never change. Assume the bound holds below rank k. The parent tuple of a rank-k site can change at most $\sum_{u \in P(v)} M(u)$ times. Before the first parent change, between consecutive parent changes, and after the last parent change, at most one effective update at v is possible on each fixed-parent interval: after that update, v equals the value prescribed by the fixed parent tuple. Hence v changes at most $M(v)$ times. Summing over sites proves termination.

At a maximal endpoint no functional update is enabled, so all generating equations hold. The endpoint has the original boundary and therefore equals the unique generated extension $E(Bx)$. □

Corollary 7.5 (Synchronous settling by dependency depth)**.** *Let $x^{(0)} = x$. In one synchronous round, keep the boundary coordinates fixed and replace every generated coordinate using the values from the previous round:*

$$x_v^{(k+1)} = \begin{cases} x_v^{(k)}, & v \in V_0, \\ F_v\big(x^{(k)}|_{P(v)}\big), & v \notin V_0. \end{cases}$$

For $0 \le k \le m$, after round k every site in $V_0 \cup \cdots \cup V_k$ agrees with $E(Bx)$. Consequently $x^{(m)} = E(Bx)$. This is a depth bound for the explicit ranked system; correspondence with a separately specified compiler or update rule requires its own theorem.

Proof. Induct on k. Boundary values are fixed. If all sites of rank at most $k-1$ are correct after round $k-1$, then every parent of a rank-k site is correct, so its round-k update evaluates the defining function at the same parent tuple as $E(Bx)$. Sites of lower rank remain correct because their parents were correct in the preceding round. □

Corollary 7.6 (Acyclic Boolean circuit compiler)**.** *Every nonempty finite acyclic Boolean circuit compiles to a ranked functional system with one $\{0,1\}$-valued site per input or gate. Inputs and constant gates form V_0, with the constant coordinates restricted to their prescribed values; every other gate is ranked by its longest source-to-gate path, has its incoming wires as parents, and uses its truth function as F_v. The extension $E(b)$ is ordinary circuit evaluation on the admissible input b. Updating all nonconstant gate sites in parallel from the preceding round reaches $E(b)$ in at most the circuit depth, independently of the initial gate-register values.*

Proof. The topological ranking makes every nonconstant gate depend only on lower ranks, so the compilation satisfies the definition on the boundary domain where constants have their prescribed values. Recursive evaluation is E, and the depth bound is Corollary 7.5. □

Corollary 7.7 (Certified solver)**.** *The procedure "settle the generating equations asynchronously, then evaluate the audits" returns either the unique consistent extension or a named failed audit. It cannot confuse an empty fiber with nontermination.*

Theorem 7.8 (Propagation of boundary perturbations)**.** *Equip every X_v with a metric d_v. Suppose that for each generated site v there are coefficients $L_{vu} \geq 0$ such that*

$$d_v(F_v(p), F_v(p')) \leq \sum_{u \in P(v)} L_{vu} d_u(p_u, p'_u).$$

For a boundary site $r \in V_0$, define coefficients recursively by

$$K_{rr} = 1, \qquad K_{vr} = 0 \ (v \in V_0,\ v \neq r), \qquad K_{vr} = \sum_{u \in P(v)} L_{vu} K_{ur} \ (v \notin V_0).$$

Then for all boundary values b, b' and every site v,

$$d_v(E(b)_v, E(b')_v) \leq \sum_{r \in V_0} K_{vr} d_r(b_r, b'_r).$$

With the ℓ^1 product metrics,

$$d_Q(E(b), E(b')) \leq \Lambda_E d_{\mathcal{B}}(b, b'), \qquad \Lambda_E := \max_{r \in V_0} \sum_{v \in V} K_{vr}.$$

Thus the generated extension supplies an explicit inverse-observation modulus $\omega_B(t) \leq \Lambda_E t$ on every accepted audit branch.

Proof. The sitewise estimate follows by induction on rank. Boundary sites satisfy it by definition. At a generated site, apply the local Lipschitz inequality and substitute the induction hypotheses; regrouping coefficients gives the recursion for K_{vr}. Summing over v and then bounding each boundary coefficient by Λ_E proves the ℓ^1 estimate. □

Remark 7.9. The local updates need not commute when one site depends on another. Order independence follows from well-founded dependency and unique extension, not from a global pairwise-commutation hypothesis. Theorem 7.8 additionally exposes the amplification carried by every dependency path.

8 Linear quotient systems and observability

Let V, W, Z be finite-dimensional real inner-product spaces. Let

$$A : V \to Z, \qquad C = \ker A, \qquad G \subseteq C$$

be a gauge subspace. Let $B : V \to W$ vanish on G, so it descends to V/G. Equip V/G with the quotient norm.

Proposition 8.1 (Linear information-boundary criterion)**.** *The induced map $C/G \to W$ is injective if and only if*

$$\ker(B|_C) = G.$$

Equivalently,

$$\operatorname{rank}(B|_C) = \dim C - \dim G.$$

The same statement holds over any field.

Proof. The kernel of the induced map is $\ker(B|_C)/G$. □

Let

$$H = C \cap G^{\perp}.$$

Under the injectivity hypothesis, $B|_H$ is injective. Let

$$\kappa_A := \|A^{\dagger}\|, \qquad \kappa_B := \|(B|_H)^{\dagger}\|,$$

where † denotes the Moore–Penrose pseudoinverse. Thus, when $H \neq \{0\}$, κ_B is the reciprocal of the smallest singular value of $B|_H$; on the zero space the pseudoinverse convention gives $\kappa_B = 0$. The constant κ_A is the global linear error-bound constant for $C = \ker A$.

Theorem 8.2 (Linear stability certificate)**.** *For all $x, y \in V$,*

$$\begin{aligned} d_{V/G}([x],[y]) \leq \kappa_A(\|Ax\| + \|Ay\|) \\ + \kappa_B(\|B(x-y)\| + \|B\|\kappa_A(\|Ax\| + \|Ay\|)) . \end{aligned} \tag{8.1}$$

Proof. Let $c_x = P_C x$ and $c_y = P_C y$ be the orthogonal projections onto C. The pseudoinverse estimate gives

$$\|x - c_x\| \leq \kappa_A \|Ax\|, \qquad \|y - c_y\| \leq \kappa_A \|Ay\|.$$

Let $h_x = P_H c_x$ and $h_y = P_H c_y$. Since $C = G \oplus H$, the vectors h_x, h_y are the minimal-norm representatives of the quotient classes $[c_x], [c_y]$. Because B vanishes on G,

$$B(h_x - h_y) = B(c_x - c_y).$$

Hence

$$\|B(h_x - h_y)\| \leq \|B(x-y)\| + \|B\|(\|x - c_x\| + \|y - c_y\|),$$

and injectivity of $B|_H$ gives

$$d_{C/G}([c_x],[c_y]) = \|h_x - h_y\| \leq \kappa_B \|B(c_x - c_y)\|.$$

Finally,

$$d_{V/G}([x],[c_x]) \leq \|x - c_x\|, \qquad d_{V/G}([y],[c_y]) \leq \|y - c_y\|,$$

so the quotient triangle inequality yields (8.1). □

The finite-field version retains the rank criterion but not singular values. It is the quotient form of an information-set condition from coding theory [29].

8.1 Finite linear dynamics

Let $T : U \to U$ be a linear update, let $S : U \to Y$ be a sensor, and let $G \subseteq U$ satisfy $T(G) \subseteq G$ and $S(G) = 0$. Equip Y^{k+1} with its product inner product. The length-k observation map is

$$\mathcal{O}_k x = (Sx, STx, \ldots, ST^k x).$$

Corollary 8.3 (Observability modulo gauge)**.** *The time tube $\mathcal{O}_k$ reconstructs U/G if and only if*

$$\ker \mathcal{O}_k = G.$$

Over $\mathbb{R}$ or $\mathbb{C}$ and when $G^\perp \neq \{0\}$, the smallest singular value of $\mathcal{O}_k$ on $G^\perp$ is the separation margin; its reciprocal (equivalently, the norm of the restricted pseudoinverse) is the inverse-stability constant. On the zero space only the pseudoinverse convention is used. The static Rule-90 example in Section 10 instead instantiates the earlier finite-field quotient rank criterion; it is not asserted to instantiate this time-tube corollary.

This is the classical observability matrix viewed through the quotient-boundary language; no novelty is claimed for the control-theoretic rank test [31].

9 Complexity of succinct boundary problems

The finite theory is elementary when Q and C are listed explicitly. It becomes computationally nontrivial when consistency is given by a Boolean circuit or formula.

Definition 9.1 (Succinct Boolean boundary instance)**.** A succinct instance consists of a Boolean circuit $F : \{0,1\}^n \to \{0,1\}$ and a coordinate set $S \subseteq \{1,\dots,n\}$. The consistent set is $C = F^{-1}(1)$ and the observation is the projection $B_S(x) = x|_S$.

Theorem 9.2 (Boundary realizability is NP-complete)**.** *Given (F, S, b), deciding whether*

$$\{x : F(x) = 1,\ B_S(x) = b\}$$

is nonempty is NP-complete.

Proof. A satisfying extension is a polynomial witness. SAT reduces to the problem by taking $S = \varnothing$. □

Theorem 9.3 (Boundary determination is coNP-complete)**.** *Given (F, S), deciding whether $B_S|_C$ is injective is coNP-complete, even when S contains all but one variable.*

Proof. Non-injectivity has an NP certificate: two distinct satisfying assignments agreeing on S. For coNP-hardness, reduce UNSAT while keeping the consistent set nonempty. Given a formula $\varphi(x_1,\dots,x_m)$, introduce bits z, t and define

$$F(x,z,t) = (\neg t \wedge [x = 0^m] \wedge \neg z) \vee (t \wedge \varphi(x)), \qquad S = \{x_1,\dots,x_m,t\}.$$

The baseline assignment $(0^m, 0, 0)$ is always consistent. If φ is satisfiable, any satisfying x gives the two distinct consistent assignments $(x, 0, 1)$ and $(x, 1, 1)$ with the same boundary. If φ is unsatisfiable, the baseline is the only consistent assignment, so the boundary restriction is injective. Thus boundary determination is equivalent to unsatisfiability even for nonempty C, and S observes all but one variable. □

Theorem 9.4 (Strong repair existence is Π_2^P-complete)**.** *Let a Boolean circuit $R(w, d)$ represent a local consistency relation on write bits w and protected collar bits d. Deciding whether an unrestricted total strong collar-preserving repair (equivalently, a set-theoretic section of the collar projection) exists is Π_2^P-complete.*

Proof. By Theorem 6.2, a strong repair exists exactly when

$$\forall d\ \exists w \quad R(w, d) = 1.$$

This is in Π_2^P and is Π_2^P-hard by the canonical quantified-Boolean-formula problem with one universal and one existential block [45, 47]. □

Remark 9.5. These results explain why an abstract finite normalizer can be mathematically canonical yet computationally difficult. The ranked functional class avoids the hardness by supplying a directed generative presentation; the linear class avoids it through rank computation.

Proposition 9.6 (Explicit-table algorithms)**.** *If Q, C, B, d_Q, $d_{\mathcal{B}}$, and Φ are given by explicit finite tables, then:*

(i) the fiber trichotomy and partial normalizer are computable in time $O(|Q| + |C|)$ after hashing observations;

(ii) ω_B is computable from $O(|C|^2)$ pairs;

(iii) η_Φ is computable from all-pairs distances to C and a sort by residual value.

10 Examples and separation results

Example 10.1 (Confluence does not imply observable reconstruction)**.** Let $Q = C = \{a, b\}$ with the empty rewrite relation and a constant observation map. Every state is a normal form and the relation is confluent, but the sole observation fiber contains two consistent states.

Example 10.2 (Two-bit witness)**.** Let $Q = \mathbb{F}_2^2$, let $B(u, v) = u$, and set $C = \{(0, 0), (1, 0)\}$. For each $u \in \mathbb{F}_2$, add the rewrite $(u, 1) \to (u, 0)$. This preserves B; the normal forms are exactly C; and every source (u, v) terminates at $(u, 0)$. Thus sources with the same observation have the same endpoint, jointly witnessing normalization, same-source confluence, and the stronger cross-source property of Theorem 3.8 for equality.

Example 10.3 (Observable determination does not imply realizability)**.** Let $C = \{c\} \subset Q$ and let $\mathcal{B} = \{0, 1\}$ with $B(c) = 0$. The restriction $B|_C$ is injective, but the fiber over 1 is empty.

Example 10.4 (Termination does not imply completeness)**.** Let $Q = \{x, n\}$, let $x \to n$, and declare $C = \{x\}$. The relation terminates, but its only normal form is not consistent.

Example 10.5 (A singleton fiber does not replace observation preservation)**.** Let $Q = \{x, c, d\}$, let $C = \{c, d\}$, set $B(x) = B(c) = 0$ and $B(d) = 1$, and take $x \to d$ as the sole rewrite step. Then the normal forms are exactly C and the consistent fiber over 0 is the singleton $\{c\}$. Nevertheless, the boundary-0 state x normalizes to d, not to c, because the rewrite changes the observation. Thus the observation-preservation hypothesis in Corollary 3.12 is load-bearing.

Example 10.6 (Native protected-profile separations)**.** The machine-checked artifact instantiates Theorem 6.7 and Proposition 6.8 on four reader-facing boundaries.

- A fine TwoBit observation reaches the final layer, while its coarse observation is almost surely reachable but has two endpoint classes; the inherited coarse rewrite remains confluent.
- A full fixed relation satisfies the static collar criterion, whereas a relation missing one collar value activates the empty-fiber cut; filling that value is a negative control.

- A proper-target three-state chain with a reachable closed trap has positive hit probability strictly below one; another hits almost surely and has two inequivalent positive endpoints.
- Two profiles reverse their strict comparison between the first and second cut coordinates, so neither is below the other in the product preorder. A separate nonidentity state map activates exact transport.

These fixtures establish nonvacuity and sharp separation. They do not claim empirical representativeness, stochastic rates, or a global mechanism rank.

10.1 Rule 90 as a regression test

No cellular automaton is used in the general theory. Rule 90 is retained because a three-cell instance simultaneously exhibits a hidden kernel, a proper information boundary, a deficient boundary, and a reverse-repair obstruction.

Over $\mathbb{F}_2$, define

$$F(a, b, c) = (b, a + c, b).$$

A one-step spacetime record is a pair (s, t) satisfying $t = F(s)$.

Proposition 10.7 (Width-three Rule 90)**.** *The kernel and image are*

$$\ker F = \operatorname{span}\{(1, 0, 1)\}, \qquad \operatorname{im} F = \{(u, v, u) : u, v \in \mathbb{F}_2\}.$$

The coordinate observation $B_{01}(u, v, w) = (u, v)$ is injective on $\operatorname{im} F$, while $B_{02}(u, v, w) = (u, w)$ is not. No total exact repair can preserve an arbitrary target row and alter only the seed.

Proof. The kernel and image follow immediately from the formula for F. On the image, the first two coordinates determine (u, v, u), whereas the first and third coordinates repeat the same bit. A reverse repair preserving every target would require F to be surjective by Theorem 6.2; for example $(0, 0, 1)$ is not in the image. □

The width-three example is a regression test, not a rule selected by the theory or a load-bearing assumption.

11 Application: set-valued public-capacity readback

The empty/singleton/ambiguous fiber distinction has a direct application to finite observer-record capacity. This subsection records the application; it does not make capacity an implication of confluence. Fix a supplied positive carrier dimension D, a trial system with terminal physical quotient fiber $\widetilde{\Omega}_{r,D}$, and a separately constructed functional

$$M_\varepsilon : \widetilde{\Omega}_{r,D} \longrightarrow \mathbb{N}_{>0}.$$

In the OPH capacity application, $M_0(q)$ is the independence number of the compound confusability graph built from reachable compatible public record atoms and globally coupled checkpoint kernels. That construction, its publicness policy, and its physical carrier are external data for the present normal-form theorem.

Define the capacity readback fiber

$$\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.$$

Proposition 11.1 (Capacity-readback trichotomy)**.** *The readback is unavailable when the functional is undefined on any required terminal class or the terminal fiber is empty, scalar when* $\mathfrak{F}_{r,\varepsilon}(D) = \{m\}$*, and ambiguous when* $|\mathfrak{F}_{r,\varepsilon}(D)| > 1$*. Stable exact closure at* D *is the singleton condition*

$$\mathfrak{F}_{r,0}(D) = \{D\}.$$

Confluence from each fixed initial state does not imply this cross-source singleton condition.

Proof. The first three statements are the image-set alternatives for the declared functional. Stable closure requires every admissible terminal source class to have full readback D. Confluence identifies repair schedules from one initial class; it does not identify distinct source classes in the unclosed fiber. □

If an implementation isomorphism preserves observer lineage, record atoms, interface atom maps, reachable public sections, publicness policy, every global checkpoint kernel, and capacity-carrier projection ranks, then it transports the compound confusability graph and every approximate decoder. The functional M_ε and the readback set are therefore invariant under that full packet isomorphism. A normal-form hash, stagewise singleton, or implementation relabeling alone supplies no such theorem. A refinement-limit statement additionally requires uniform stable separation or an explicit capacity-packet naturality receipt, in accord with the general refinement results above.

12 Discussion

12.1 Why the factorization is useful

The estimate

$$\text{residual} \xrightarrow{\eta} \text{distance to consistency} \xrightarrow{\omega} \text{observable uniqueness}$$

separates two burdens that are often mixed. A rewrite algorithm, numerical solver, or physical implementation controls η by showing that a small residual means closeness to C. The observation architecture controls ω by showing that $B|_C$ is well conditioned, a property strictly stronger than injectivity of $B|_C$. Critical-pair arguments may help establish exact completion or small residuals, and they are not required in the final output comparison.

12.2 Why finite uniqueness is not enough

In a refinement program, proving $B_r|_{C_r}$ injective for every r does not prevent the inverse condition number from diverging. Example 4.15 is deliberately minimal. The correct limit-facing statement is uniform stable separation, not a list of stagewise singleton claims. This is the main stability requirement emphasized for refinement-sensitive applications.

The protected first-hit profile has the same discipline at a different interface. Finite hitting, closed-class, Bellman, and lumpability arguments are standard; the OPH-specific step is to assemble them around protected observations and declared sources, recover the existing behavior cuts, and retain every source, target, quotient, and scheduler premise during transport.

12.3 A declared scheduler class and its capacity split

The machine-checked development fixes one declared admissible scheduler class on the two-bit repair carrier. A scheduler in the class is a finite Markov kernel whose supported steps do not increase a

declared defect mismatch, write at most one of the two registers, and spend at most one unit of expected declared capacity per step. Reading the observed register as the shared network record and the defect flag as a private register splits each step cost into a shared and a private part and cuts out two sub-classes: the self-update-only schedulers, which never write the shared register, and the network-update-only schedulers, which never write the private one. Three results hold on this carrier. Under every scheduler in the class the defective fiber is unreachable from every consistent state, while an explicit mismatch-violating kernel outside the class reaches it with probability one. The fiber with the shared register flipped is unreachable under every self-update-only scheduler and is hit with probability one by an explicit network-update-only member, so a positive hit prices at least one shared write. The almost-sure hitting hypothesis of the fine two-bit fixture is discharged by the declared repair kernel, and the in-class identity scheduler witnesses that the class constraints do not force it. The class is a declaration on this finite carrier: the locality and capacity clauses are typed and unconsumed by the obstruction theorems, and no scheduler-selection claim is made.

The endpoint-uniqueness cut is quantified the same way on the coarse fixture, whose target admits both shared values and whose two declared sources carry a raised flag under each shared value. No self-update-only scheduler places the coarse behavior in L_3: with positive hits from both sources, each endpoint support keeps the shared value of its source, so the two supports differ. No network-update-only scheduler hits the coarse target from a raised flag, since such a scheduler never clears the flag. An explicit merge-then-clear class member, which writes the shared register on one supported step and clears the private flag on another and so lies in neither sub-class, hits from both sources with probability one and places the coarse behavior in L_3. Every in-class scheduler that does so spends both capacities on supported steps. On the fine fixture the target fiber is a singleton, so $L_2 = L_3$ under every scheduler in the class. Each of the four cuts therefore carries a statement quantified over the class or over a sub-class on this carrier.

12.4 Adaptive canonical repair: stutter, fairness, and endpoint promotion

The declared finite Markov profile above and the repository's canonical adaptive repair answer different questions. Let C be a finite OPH carrier, let σ choose one site from the active record at each attempt, and write

$$x_n = \mathrm{adaptiveRun}_C(n, \sigma, x_0), \qquad r_n = \mathrm{mismatchCount}(x_n).$$

An attempt may stutter when the chosen site has no effective repair. Every nonstuttering attempt is an accepted repair step and strictly lowers r_n, while every stuttering attempt leaves the record unchanged.

Theorem 12.1 (Adaptive repair stratification)**.** *For every x_0 and every adaptive site scheduler σ, there is an N such that $x_n = x_N$ for all $n \geq N$. If σ is pathwise weakly fair, meaning that a site which remains enabled along a tail is eventually selected on that tail, then x_N is a normal form. Under the stronger work-conserving condition, which requires the selected site to fire whenever some site can fire, one may choose $N \leq \mathrm{mismatchCount}(x_0)$, and the run is constant from N onward.*

Proof. The mismatch count never increases and strictly decreases at every genuine change, so only finitely many genuine changes are possible. If the eventual constant state were reducible, one site would remain enabled throughout the constant tail; pathwise weak fairness would select it and force another genuine change, a contradiction. Under work conservation every nonnormal state takes a genuine decreasing step, so the initial natural-number rank bounds the first normal index. □

The machine-checked provider also proves that every x_n is reachable from x_0 by accepted repair steps. A downstream endpoint theorem then adds the separately named completeness and confluence hypotheses: every pathwise weak-fair adaptive run reaches the same raw endpoint as the canonical repair, and therefore the same public fixed-point object. This promotion supplies no stochastic kernel, active-source law, hitting probability, rate, physical scheduler, or refinement theorem.

12.5 Cumulative attempt capacity and sharpness

Every scheduler invocation receives one attempt unit, including an equality stutter. Let $A_N = N$ be this cumulative attempt cost, let G_N count the genuine changes among the first N attempts, and let $S_N = N - G_N$. Thus $A_N = G_N + S_N$; the initial mismatch rank controls G_N, but it does not control S_N.

Definition 12.2 (Bounded waste)**.** For $q \in \mathbb{N}$, an adaptive scheduler has *waste bound* q if, from every scheduler index and every reducible record, at least one genuine change occurs among the next $q + 1$ attempts. Work conservation is the special case $q = 0$.

Theorem 12.3 (Cumulative attempt-capacity classification)**.** *The canonical OPH adaptive attempt semantics has the following three properties.*

(i) There is no single mismatch-only finite attempt horizon uniformly valid for all normalizing schedulers. For every d, a pathwise weak-fair scheduler on the committed TwoCell source has initial mismatch one, stutters through attempt d, and reaches a normal form at attempt $d+1$.

(ii) A scheduler with waste bound q reaches a stable normal form at some

$$N \le (q+1)\,\mathrm{mismatchCount}(x_0).$$

Consequently an attempt budget B at least this large cannot be exhausted before normality.

(iii) The work-conserving coefficient is sharp. For every $m \ge 1$, a canonical independent-defect carrier with m self-loop mismatches has a work-conserving scheduler whose first normal index is exactly m. Budget below m fails and budget at least m succeeds. The committed TwoCell source is a second sharp instance with a different patch cardinality.

Proof. For (i), the delayed scheduler selects the edge-free TwoCell site for its first d indices and selects the enabled probe next. All delayed attempts are literal equality stutters, while the final probe repair is genuine. For (ii), if the current record is reducible, bounded waste produces a genuine change within $q + 1$ attempts. That change strictly lowers the natural mismatch rank. Strong induction on the rank gives the displayed product bound, and normality makes the remaining tail constant. For (iii), each site of the independent-defect carrier owns one self-loop mismatch. A canonical local repair sets exactly that site to its unique consistent value, so every genuine change lowers the mismatch count by exactly one. The state-dependent scheduler chooses a broken site whenever one exists. Its rank after n attempts is $m - n$ for $n \le m$, proving both necessity and equality. The TwoCell calculation gives the second, nonisomorphic threshold-one instance. □

Under the same budget and waste premises, the downstream Tower theorem adds the separately named completeness and confluence hypotheses and returns the existing canonical raw and public fixed-point endpoint at an index $N \le B$. The capacity is a mathematical count of scheduler invocations. It is not a physical clock, rate, energy, bandwidth allocation, hardware quota, fee, or stochastic policy.

12.6 Fixed computation federations

The computation compiler has a second, distinct asynchronous surface. For a Boolean formula φ on k inputs, the generated node list $\mathrm{go}(\varphi, k)$ is independent of the input. Its registers start at k; the initial state supplies the lower input ports. A canonical accepted step chooses a failing node and performs exactly that node's declared write. The historical patch-frame relation remains unchanged and is not used as the accepted relation.

Read the node defects in increasing register order, with every earlier bit more significant than all later bits together. This superincreasing natural rank strictly decreases at every canonical accepted step. An upstream repair may expose several later defects: the machine-checked fanout control raises the raw failing-node count from one to two, so ordinary defect count is not the measure. The dependency rank nevertheless falls, and the accepted relation is well founded without a fairness premise.

Theorem 12.4 (Fixed computation-federation progress)**.** *For every finite Boolean function there is one input-independent generated-node federation such that every canonical attempt run is eventually constant. If every node that remains failing on a tail is eventually selected on that tail, the stable state is a consensus and its output equals the function value for the preserved initial input. Any two normal endpoints with the same input observation agree on the output register.*

The fairness premise removes attempt stutter; it does not prove accepted-step termination. Its positive singleton instance and a root-only starvation schedule inhabit the two sides. The broader historical relation admits a sharper one-member control: the only member and site are selected at every attempt and remain continuously enabled, yet the run stutters forever outside consensus because the relation does not require the declared repair to execute. Thus fairness cannot compensate for a step semantics that counts a no-op.

Theorem 12.5 (Fixed-federation execution classification)**.** *For a well-formed generated-node federation and a fixed attempt run, pathwise weak fairness holds if and only if the run eventually reaches a stable consensus. The reverse implication is vacuous after consensus: no node can remain failing on that stable tail, so it does not imply recurrent selection. At the compiler's present node granularity, member recurrence and register-site recurrence coincide because generated output registers are unique; recurrence is nevertheless strictly stronger than tail fairness. A state-blind round-robin scheduler is member- and site-recurrent and reaches the stable correct output. More sharply, its first n attempts are exactly the emission-order sweep, so it reaches stable consensus and the correct output within one emitted-node cycle.*

If the emitted program has n nodes, every canonical genuine accepted path has length at most

$$\frac{n(n+1)}{2}.$$

An actual recursively nested NAND family emitted by the same compiler has a consensus-reaching accepted path of length m satisfying

$$8m + 1 = n^2 + 8n.$$

Thus the accepted-step classification is order-sharp $\Theta(n^2)$ in one node-count metric; the finite upper and lower constants are not claimed equal. Well-formedness alone does not yield this: every well-formed node list admits fewer than 2^n canonical accepted steps, an explicit fanout chain family attains $2^n - 1$, and at three nodes that family realizes seven accepted steps against the triangular budget six, so the compiler's single-consumer structure is what collapses the exponential regime to a

quadratic one. Weak fairness alone supplies no uniform finite attempt horizon. If a supplied fixed-node scheduler instead has bounded waste q, stable consensus is reached within $(q+1)n(n+1)/2$ attempts. Round robin discharges this premise with $q = n-1$, but its direct one-cycle theorem gives the sharper n-attempt bound for that scheduler. The Tower consumer returns the same-input unique output.

The bounded-waste counting theorem is shared with the canonical adaptive repair route through a neutral ranked-attempt interface. Each domain proves its own run and quiescence equivalence, and each Tower receiver calls the shared theorem. This is code and proof reuse, not a claim that adaptive scheduler waste and fixed-compiler repair work are the same scientific quantity.

The endpoint statement uses the immutable input prefix as the protected observation and output-register equality as the endpoint equivalence. Unused natural-number registers are not identified. The selected round-robin object is a mathematical function, not a physical scheduler source. No physical device, finite `PublicWorld` presentation, clock, rate, energy, bandwidth, fee, hardware realization, or raw-state uniqueness is constructed.

12.7 A settled public class with fluctuating spatial readback

The finite conservative port-repair family separates quotient normality from quiescence of a richer observable. Its balanced integer sector has r raised loads among p ports. Every arrangement has the same component multiset, while fair repair on a connected seam graph exchanges unequal adjacent loads with probability $1/2$. The quotient readback is fixed, but spatial arrangements and their histories can continue to change. Consequently, uniqueness of this multiset does not identify a unique spatial configuration or a law for a spatial observer readout.

For the centered load $z = n - (r/p)\mathbf{1}$, state-independent weighted Poisson attempts give a reversible exclusion chain with

$$Gz = -\tfrac{1}{2}Lz, \qquad \mathrm{Cov}_\pi(z) = \frac{r(p-r)}{p(p-1)}\left(I - \frac{1}{p}\mathbf{1}\mathbf{1}^{\mathsf{T}}\right).$$

Write $\kappa = r(p-r)/[p(p-1)]$. Uniformity on fixed-occupancy configurations follows from symmetric swap rates, and graph connectedness gives irreducibility. Thus distinct stationary readouts of the same repair system can have distinct spectra. For example, the long-history fluctuation of z has covariance $4\kappa L^{+}$, whereas that of Lz has covariance $4\kappa L$. These are different observations of one chain, not conflicting normal forms. The finite rational four-port, six-state instance in `Lean/Thermodynamics/NativeRepairFourCycle.lean` certifies generator, stationary, drift, Poisson and covariance identities, including the load/drive distinction. The general fixed-graph stochastic limit is an analytical theorem of the screen-microphysics companion, not a consequence of a normal-form theorem alone.

The large retained configurations give an empirical version of the same distinction. There are sixteen schedules at each of levels eight through ten, with one common initial load field per level. Consistently centered ancestral-group readbacks split terminal power exactly into unbiased conditional schedule variance and cross-schedule common power. At level ten the initial-to-terminal linear response is 0.05643 for individual carrier groups and 0.98983 for 65,536-carrier groups. The latter schedule-noise fraction is 0.143%; its terminal variance is 11.048 times the stationary balanced null. Coarse spatial information therefore survives in these end-of-settling-sweep configurations even when their public multisets agree. The observations are conditional on the fixed initial fields and stopping rule. They do not make the public multiset spatially complete or establish an asymptotic transport law. The estimator, receipt pins and independent controls are preserved under `eviden ce/observer_dynamics_20260925/sim-analysis/`.

These examples concern observer-like bounded patches with local state, ports, record readback and repair. Their lesson for observation design is precise: invariance of one published quotient does not determine the covariance, memory, geometry or clock of an additional readout. Each extra observable requires its own factorization or dynamical argument.

12.8 Limitations

The theory assumes a declared meaningful quotient, consistent set, residual, observation metric, and refinement map. It does not select those objects. In particular:

- a normalizer is not a probability measure;
- an inverse-limit normalizer is not a continuum field theory;
- a finite update bound is not a wall-clock liveness theorem;
- a boundary pseudometric does not acquire additional semantics without a separate interpretation theorem;
- a small Rule-90 example does not establish cellular-automaton universality or generality.
- a finite protected profile is relative to one declared scheduler and active-source family; it gives no rate, expected hitting time, mixing bound, stationary law, infinite-state extension, or refinement-tower conclusion; results quantified over the declared admissible class are relative to that class declaration and to the two-register capacity-split reading;
- canonical adaptive repair is eventually constant without a fairness premise, but normality requires pathwise weak fairness and a canonical consistent endpoint requires completeness and confluence;
- cumulative attempt capacity counts every chosen-site invocation, including equality stutter; bounded waste and the budget are supplied premises, weak fairness alone gives no uniform mismatch-only finite attempt bound, and the sharp lower bound is family-specific rather than universal over all carriers;
- fixed-federation pathwise weak fairness is exactly stable consensus only relative to one attempt run and the source-derived eventual-constancy theorem; its reverse is vacuous after stable consensus and does not give recurrence, weak fairness alone has no uniform finite attempt horizon, the accepted-step result is order-sharp rather than constant-sharp, and bounded-waste attempts have no physical unit;
- coordinatewise cut comparison is not a scalar score, total order, or family-wide optimum, and exact finite transport is not implementation or product refinement.

These exclusions are necessary for mathematical correctness.

13 Data availability

The repository contains the PDF, LaTeX source, bibliography, reproducible build receipt, archive manifest, and SHA-256 checksums. The mathematical proofs in this manuscript are self-contained. No empirical dataset is used.

14 Conclusion

Confluence is about paths from one state. Observable normalization is about what partial data determine after consistency is enforced. Their exact intersection is classical unique-normal-form theory: under observation preservation and $\mathrm{NF}(\to) = C$, identification modulo any silent equivalence is equivalent to agreement of normal endpoints across all sources with the same observation. This does not by itself provide normalization or all-schedule settlement. Their stable intersection is governed by two analytic objects: an error bound to the consistent set and an inverse modulus for the observation map on that set.

On the canonical OPH adaptive branch, finite genuine-change stabilization is source-derived rather than postulated as a scheduler property. The remaining steps are deliberately separate: pathwise weak fairness supplies normality, while completeness and confluence promote the stable state to one consistent canonical raw and public endpoint.

On the separate fixed compiler branch, the emission-order round-robin schedule reaches the exact output in one n-attempt cycle, arbitrary genuine accepted order has an order-sharp $\Theta(n^2)$ worst-case family, weak fairness alone has no uniform finite attempt horizon, and bounded waste supplies the general conditional attempt bound. These four statements classify distinct scheduler and cost regimes rather than attaching a physical duration to an attempt.

The resulting estimate is universal and sharp in its two unavoidable components. It survives refinement: one-step defects telescope, same-level solver agreement needs no tower term, and compatible cross-level computations differ by their receipt errors and the intervening normalizer-defect tail. For a compatible admissible input in complete spaces, nonexpansive restrictions, vanishing receipt errors, and summable pathwise tails give a common projective limit; exact compatibility recovers the inverse-limit normalizer. On finite probability spaces, fiber averaging supplies the canonical conditional-expectation projector, while its matrix receipt keeps that averaging distinct from representative selection or a spectral-gap claim. Empty fibers and ambiguous fibers stay visible; a repair operator does not absorb them. The theory therefore supplies the determination and stability question, a four-stage protected-behavior ladder, and an orthogonal local-support question:

Does a consistent extension exist?
Is it observably unique and stable?
Does some declared source reach it with positive probability?
Do all declared sources reach it almost surely?
Do first-hit endpoints select one silent-equivalence class?
Can the declared local support construct it?

No one of these questions answers the others. Keeping them separate is the central mathematical lesson and the reason the framework applies across rewrite, reconstruction, and distributed-inference models.

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Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Finite Markov first-hit theory, qualitative reachability, closed classes, strong lumpability, and product preorders [9, 17, 33].
- Newman's lemma, abstract rewriting, and observability of consistent states [8, 43].
- Scheduler-quantified reachability for finite Markov decision processes and probabilistic model checking [9, 11].
- Stability moduli, metric subregularity, approximate inverse systems, and telescoping bounds [19, 40].
- Weighted abstract reduction systems, derivational complexity, amortized potential bounds, fairness, and self-stabilization [18, 22, 26, 48, 50].
- Well-founded descent and weak fairness in finite asynchronous rewriting [8, 22].

References

[1] Samson Abramsky. A generalized kahn principle for abstract asynchronous networks. In *Mathematical Foundations of Programming Semantics*, volume 442 of *Lecture Notes in Computer Science*, pages 1–21, 1990. doi: 10.1007/BFb0040252.

[2] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011. doi: 10.1088/1367-2630/13/11/113036.

[3] Takahito Aoto and Yoshihito Toyama. Automated proofs of unique normal forms w.r.t. conversion for term rewriting systems. In *Frontiers of Combining Systems (FroCoS 2019)*, volume

11715 of *Lecture Notes in Computer Science*, pages 330–347, 2019. doi: 10.1007/978-3-030-29007-8_19.

[4] Julio Aracena, Maximilien Gadouleau, Adrien Richard, and Lilian Salinas. Fixing monotone boolean networks asynchronously. *arXiv preprint arXiv:1802.02068*, 2019. URL https://arxiv.org/abs/1802.02068.

[5] Marcelo Arenas, Leopoldo Bertossi, and Jan Chomicki. Consistent query answers in inconsistent databases. In *Proceedings of the Eighteenth ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, pages 68–79, 1999. doi: 10.1145/303976.303983.

[6] Albert Atserias and Phokion G. Kolaitis. Consistency, acyclicity, and positive semirings. In *Samson Abramsky on Logic and Structure in Computer Science and Beyond*, pages 623–668. Springer, 2023. doi: 10.1007/978-3-031-24117-8_17.

[7] Martin Avanzini and Akihisa Yamada. Weighted rewriting. In *Frontiers of Combining Systems*, volume 15979 of *Lecture Notes in Computer Science*, pages 191–208, 2025. doi: 10.1007/978-3-032-04167-8_11.

[8] Franz Baader and Tobias Nipkow. *Term Rewriting and All That*. Cambridge University Press, 1998. doi: 10.1017/CBO9781139172752.

[9] Christel Baier and Joost-Pieter Katoen. *Principles of Model Checking*. MIT Press, 2008. ISBN 978-0-262-02649-9.

[10] Catriel Beeri, Ronald Fagin, David Maier, and Mihalis Yannakakis. On the desirability of acyclic database schemes. *Journal of the ACM*, 30(3):479–513, 1983. doi: 10.1145/2402.322389.

[11] Andrea Bianco and Luca de Alfaro. Model checking of probabilistic and nondeterministic systems. In *Foundations of Software Technology and Theoretical Computer Science (FSTTCS 1995)*, volume 1026, pages 499–513. Springer Berlin Heidelberg, 1995. doi: 10.1007/3-540-60692-0_70.

[12] Graham Campbell and Detlef Plump. Confluence up to garbage in graph transformation. *Theoretical Computer Science*, 884:1–22, 2021. doi: 10.1016/j.tcs.2021.06.010.

[13] Daizhan Cheng, Zhiqiang Li, and Hongsheng Qi. Realization of boolean control networks. *Automatica*, 46(1):62–69, 2010. doi: 10.1016/j.automatica.2009.10.036.

[14] Henning Christiansen and Maja H. Kirkeby. On proving confluence modulo equivalence for constraint handling rules. *Formal Aspects of Computing*, 29:57–95, 2017. doi: 10.1007/s00165-016-0396-9.

[15] Radek Cibulka, Asen L. Dontchev, and Alexander Y. Kruger. Strong metric subregularity of mappings in variational analysis and optimization. *Journal of Mathematical Analysis and Applications*, 457:1247–1282, 2018. doi: 10.1016/j.jmaa.2016.11.045.

[16] Victor Dalmau and Jakub Opršal. Local consistency as a reduction between constraint satisfaction problems. In *Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science*, pages 1–15, 2024. doi: 10.1145/3661814.3662068.

[17] B. A. Davey and H. A. Priestley. *Introduction to Lattices and Order*. Cambridge University Press, 2002. doi: 10.1017/CBO9780511809088.

[18] Edsger W. Dijkstra. Self-stabilizing systems in spite of distributed control. *Communications of the ACM*, 17(11):643–644, 1974. doi: 10.1145/361179.361202.

[19] Asen L. Dontchev and R. Tyrrell Rockafellar. *Implicit Functions and Solution Mappings: A View from Variational Analysis*. Springer, 2 edition, 2014. doi: 10.1007/978-1-4939-1037-3.

[20] Swan Dubois and Sébastien Tixeuil. A taxonomy of daemons in self-stabilization. *arXiv preprint arXiv:1110.0334*, 2011. URL https://arxiv.org/abs/1110.0334.

[21] Dominique Duval, Rachid Echahed, and Frédéric Prost. Categorical abstract rewriting systems and functoriality of graph transformation. *Electronic Communications of the EASST*, 41, 2011. doi: 10.14279/tuj.eceasst.41.612.

[22] Nissim Francez. *Fairness*. Springer, 1986. doi: 10.1007/978-1-4612-4886-6.

[23] Maximilien Gadouleau and Adrien Richard. On fixable families of boolean networks. *arXiv preprint arXiv:1804.01931*, 2018. URL https://arxiv.org/abs/1804.01931.

[24] Daniel Gall and Thom Frühwirth. Confluence modulo equivalence with invariants in constraint handling rules. In *Functional and Logic Programming (FLOPS 2018)*, volume 10818 of *Lecture Notes in Computer Science*, pages 116–131, 2018. doi: 10.1007/978-3-319-90686-7_8.

[25] Francesco Gavazzo and Cecilia Di Florio. Quantitative and metric rewriting: Abstract, non-expansive, and graded systems. *arXiv preprint arXiv:2206.13610*, 2022. doi: 10.48550/arXiv.2206.13610.

[26] Dieter Hofbauer and Clemens Lautemann. Termination proofs and the length of derivations. In *Rewriting Techniques and Applications (RTA-89)*, volume 355, pages 167–177. Springer Berlin Heidelberg, 1989. doi: 10.1007/3-540-51081-8_107.

[27] Alan J. Hoffman. On approximate solutions of systems of linear inequalities. *Journal of Research of the National Bureau of Standards*, 49(4):263–265, 1952. doi: 10.6028/jres.049.027.

[28] Gérard Huet. Confluent reductions: Abstract properties and applications to term rewriting systems. *Journal of the ACM*, 27(4):797–821, 1980. doi: 10.1145/322217.322230.

[29] W. Cary Huffman and Vera Pless. *Fundamentals of Error-Correcting Codes*. Cambridge University Press, 2003. doi: 10.1017/CBO9780511807077.

[30] Gilles Kahn. The semantics of a simple language for parallel programming. In Jack L. Rosenfeld, editor, *Information Processing 74*, pages 471–475, Amsterdam, 1974. North-Holland.

[31] Thomas Kailath. *Linear Systems*. Prentice-Hall, 1980.

[32] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer-Verlag, New York, 1976. ISBN 978-0-387-90192-3.

[33] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics), 1976. ISBN 0-387-90192-2.

[34] Cynthia Kop, Aart Middeldorp, and Christian Sternagel. Complexity of conditional term rewriting. *Logical Methods in Computer Science*, 13(1):1–56, 2017. doi: 10.23638/LMCS-13(1:6)2017.

[35] Costas Kravaris. On functional observability of nonlinear systems and the design of functional observers with assignable error dynamics. *IFAC-PapersOnLine*, 59(19):668–673, 2025. doi: 10.1016/j.ifacol.2025.11.112.

[36] Alexander Y. Kruger. Error bounds and metric subregularity. *Optimization*, 64(1):49–79, 2015. doi: 10.1080/02331934.2014.938074.

[37] Leslie Lamport. Fairness and hyperfairness. *Distributed Computing*, 13(4):239–245, 2000.

[38] Salvador Lucas. Confluence of conditional rewriting modulo. *arXiv preprint arXiv:2504.01847*, 2025. doi: 10.48550/arXiv.2504.01847.

[39] Sibe Mardešić. On approximate inverse systems and resolutions. *Fundamenta Mathematicae*, 142(3):241–255, 1993. doi: 10.4064/fm-142-3-241-255.

[40] Sibe Mardešić and Leonard R. Rubin. Approximate inverse systems of compacta and covering dimension. *Pacific Journal of Mathematics*, 138(1):129–144, 1989. doi: 10.2140/pjm.1989.138.129.

[41] Sibe Mardešić and Jack Segal. Stability of almost commutative inverse systems of compacta. *Topology and its Applications*, 31(3):285–299, 1989. doi: 10.1016/0166-8641(89)90025-4.

[42] Arthur N. Montanari, Leandro Freitas, Daniele Proverbio, and Jorge Gonçalves. Functional observability and subspace reconstruction in nonlinear systems. *Physical Review Research*, 4 (4):043195, 2022. doi: 10.1103/PhysRevResearch.4.043195.

[43] M. H. A. Newman. On theories with a combinatorial definition of equivalence. *Annals of Mathematics*, 43(2):223–243, 1942. doi: 10.2307/1968867.

[44] J. R. Norris. *Markov Chains*. Cambridge University Press, 1997. doi: 10.1017/CBO9780511810633.

[45] Christos H. Papadimitriou. *Computational Complexity*. Addison-Wesley, 1994.

[46] L. C. G. Rogers and J. W. Pitman. Markov functions. *The Annals of Probability*, 9(4):573–582, 1981. doi: 10.1214/aop/1176994363.

[47] Larry J. Stockmeyer. The polynomial-time hierarchy. *Theoretical Computer Science*, 3(1): 1–22, 1976. doi: 10.1016/0304-3975(76)90061-X.

[48] Robert Endre Tarjan. Amortized computational complexity. *SIAM Journal on Algebraic Discrete Methods*, 6(2):306–318, 1985. doi: 10.1137/0606031.

[49] Terese. *Term Rewriting Systems*, volume 55 of *Cambridge Tracts in Theoretical Computer Science*. Cambridge University Press, 2003. doi: 10.1017/CBO9780511541792.

[50] Terese. *Term Rewriting Systems*. Cambridge University Press, 2003. ISBN 0-521-39115-6.

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